

Survey of Trustworthy Representation Learning Across Domains

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As AI systems have obtained significant performance to be deployed widely in our daily lives and human society, people both enjoy the benefits brought by these technologies and suffer many social issues induced by these systems. To make AI systems good enough and trustworthy, plenty of researches have been done to build guidelines for trustworthy AI systems. Machine learning is one of the most important parts of AI systems, and representation learning is the fundamental technology in machine learning. How to make representation learning trustworthy in real-world application, e.g., cross domain scenarios, is very valuable and necessary for both machine learning and AI system fields. Inspired by the concepts in trustworthy AI, we proposed the first trustworthy representation learning across domains framework, which includes four concepts, i.e., robustness, privacy, fairness, and explainability, to give a comprehensive literature review on this research direction. Specifically, we first introduce the details of the proposed trustworthy framework for representation learning across domains. Second, we provide basic notions and comprehensively summarize existing methods for the trustworthy framework from four concepts. Finally, we conclude this survey with insights and discussions on future research directions.

CCS Concepts: • **Computing methodologies** → **Computer vision representations; Learning under covariate shift; Transfer learning**; • **Security and privacy** → Privacy protections;

Additional Key Words and Phrases: Trustworthy, representation learning, cross-domain learning, robustness, fairness, privacy, explainability

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1 INTRODUCTION

Representation learning stands as a cornerstone challenge within the realm of machine learning. Its primary purpose is to encapsulate and delineate data in a manner conducive to the application of machine learning techniques. The efficacy of machine learning algorithms is profoundly contingent upon the caliber of data representation. Invariably, the potency of these algorithms is closely intertwined with the discriminative nature of the representations learned. Notably, representations endowed with heightened discriminative attributes consistently yield superior performance in contrast to their less-discriminative counterparts. As a consequence, much effort has been devoted to featuring engineering works that aim to design a good feature representation from raw data that can benefit machine learning. Early feature engineering works are highly dependent on domain-specific and human ingenuity. For example, in computer vision, some hand-designed feature descriptors, e.g., Local Binary Pattern [69], Scale Invariant Feature Transform [143], and Histogram of Oriented Gradients [34], are proposed to extract the representations of the data for recognition- and detection-related tasks. However, these well-designed descriptors heavily rely on human effort and domain specialist knowledge. To reduce the dependence on feature engineering, an automatic representation learning strategy, i.e., deep learning, is proposed to reduce the need of human labor and the requirement of specific expert knowledge. Deep learning [68, 168] has emerged as an omnipresent facet of representation learning, finding extensive application across a plethora of real-world domains, including but not limited to image classification, video captioning, and social network analysis. The confluence of burgeoning data availability and remarkable advancements in high-performance computing resources has catalyzed the proliferation of deep learning frameworks. Noteworthy examples encompass **Convolutional Neural Network (CNN)** [107], **Recurrent Neural Network** [91], **Graph Neural Network** [85], **Generative Adversarial Network (GAN)** [59], and **Transformer** [219]. This lineage of deep learning frameworks has charted remarkable strides in contrast to earlier paradigms of feature engineering. Their prowess spans a spectrum of domains, including computer vision, natural language processing, speech, audio processing, and beyond. The amalgamation of these frameworks with vast and diverse domains has culminated in substantial advancements and accomplishments.

Despite the impressive achievements exhibited by these sophisticated representation learning methodologies on extensive datasets across an array of tasks [94, 131, 132, 290], their effectiveness often falters when confronted with the complexities of real-world situations. A glaring example of this lies in their inability to seamlessly adapt to uncharted territories, such as novel datasets or unfamiliar environments. A pertinent illustration is that of a human detector fine-tuned on indoor images, which proves inadequate when exposed to testing images derived from outdoor contexts. This predicament can be attributed to the concept of domain shift [224, 225, 279], wherein a discernible disparity emerges between the distribution of the training data and that of the testing data. Even slight modifications in the training data can lead to significant performance degradation in numerous deep learning methods [72, 161, 253, 293] when applied to a target dataset. To enhance the performance of representation learning across domains in real-world scenarios, a multitude of techniques and architectures have been proposed. These approaches aim to tackle the challenges from various angles, collectively striving to address the impact of diverse factors. For instance, in the field of domain adaptation [225], learning the representations that can reduce the domain gap between source and target is crucial for models to improve the performance [139, 140, 178, 216, 289, 291, 292]; in the context of domain generalization problem [224, 279], it is very important to learn the generalizable representations to improve the robustness of the systems under different application scenarios; with regard to cross-domain few-shot problem, how to enrich the diversity of the support set is one of the pivotal direction to learn the reliable

representation to benefit downstream tasks. Inspired by the distinct demands posed by various problems in representation learning across domains, researchers are dedicating increased attention and diligence toward rendering this process more dependable, resilient, equitable, and ethically sound. The objective is to instill trust across diverse domains. In the context of this article, our aim is to deliver a timely and comprehensive survey of recent endeavors that confront the multifaceted challenges associated with establishing “trustworthiness” in representation learning across domains.

1.1 Trustworthy for AI and Representation Learning Across Domains

In recent years, **Artificial Intelligence (AI)** has deeply reshaped our daily lives and society. Various AI systems have been deployed in every corner of our life, including transportation, education, public health, security, business, and so forth. We enjoy the economic and social benefits provided AI but also face the challenges come from AI. Because of the complex system design in AI, massive demand for data, the bottleneck of computing power, and the favor of accuracy-only, the AI systems is not as reliable as expected. On the contrary, AI has induced plenty of challenges, including individual and social problems, and reduced the trust of people on it directly or indirectly. For example, many facial recognition systems suffer from the higher error rates for African-Americans, women, and young people . Another example is the AI systems used in U.S. hospitals to predict which patients would likely need extra medical care heavily favored white patients over black patients. Therefore, many organizations [1, 30] and researchers [95, 111, 122] have put much effort to proposed guidelines on how to build a more reliable AI system from different stages of the lifecycle. These endeavors lead the requirement for AI system switch from accuracy to trustworthiness.

Drawing inspiration from the achievements of trustworthy AI, the trustworthiness of representation learning across various domains is contingent upon its alignment with the fundamental principles inherent in trustworthy AI. As a result, we are convinced of the necessity for a meticulous endeavor to translate the tenets of trustworthy AI into the realm of representation learning across diverse domains. This endeavor seeks to establish a comprehensive framework that harmonizes these principles. Figure 1 provides a visual depiction of this framework, highlighting the central components: robustness, privacy, fairness, and explainability.

1.2 Concepts in Trustworthy Representation Learning Across Domains

As the trustworthy representation learning across domains is one of the specific cases in trustworthy AI, it inherits the conventional characteristics from trustworthy AI but possesses unique properties. To better illustrate our proposed trustworthy representation learning across domains framework, we will give a brief review of each concept in the framework, including concept definition and research challenges. Please note that these concepts are not orthogonal but closely correlated.

1.2.1 Robustness. In the realm of trustworthy AI, robustness entails endowing AI systems with the capability to handle errors—be it data inaccuracies, attacks, or unanticipated inputs—during decision-making, thereby yielding dependable outcomes. This is a foundational concept underpinning the application of trustworthy AI in practical scenarios. The absence of robustness renders an AI system susceptible, where even minor input perturbations can lead to starkly different results. A prime example is the case of an AI system incorrectly classifying a panda image as a gibbon with high confidence due to minute, carefully designed noise [60]. Within the domain of representation learning across diverse data sources, robustness assumes a distinct connotation. Here, it signifies the capacity of data representations to authentically and accurately depict the underlying distribution spanning different domains. While it is impractical to account for all possible

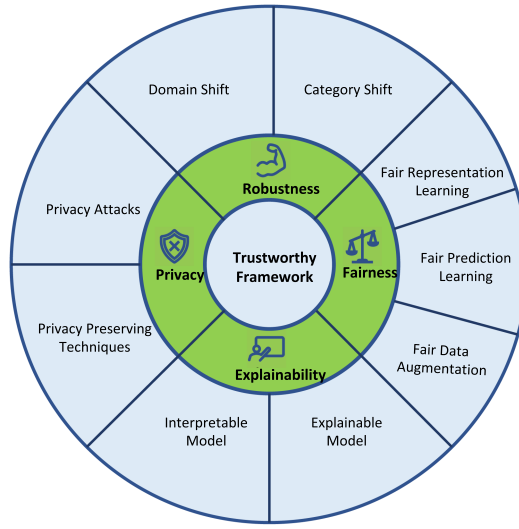


Fig. 1. Illustration of our proposed trustworthy framework, which includes four concepts, i.e., robustness, privacy, fairness, and explainability.

vulnerabilities, our focus chiefly centers on algorithms that facilitate learning data representations enabling consistent performance across domains. Therefore, they could guarantee that AI systems remain robust and keep the trustworthy representation learning intact.

1.2.2 Privacy. The objective of privacy is to safeguard the data and model within an AI system from unauthorized exposure. Given that data can encompass sensitive details like familial ties, credit histories, and medical records, the risk of unauthorized access leading to various complications is considerable. In representation learning across domains, the emphasis is on preserving data privacy. In contrast to single-domain scenarios, the exchange of information between multiple domains heightens privacy concerns. The intricate challenge lies in ensuring the security of data when dealing with diverse domains, where the potential for data leakage becomes more pronounced.

1.2.3 Fairness. Fairness demands that AI systems exhibit equality, impartiality, and lack of bias toward individuals or groups with varying attributes when making decisions or predictions. Unfortunately, the rapid proliferation of AI systems in our daily routines has exposed instances of human-like discriminatory biases and unjust decisions. For instance, the **Association for Computing Machinery (ACM)** has advocated for discontinuing the use of facial recognition technologies in both private and government spheres due to glaring biases tied to human characteristics like ethnicity, race, and gender. Additionally, Microsoft's AI-based chatbot on Twitter generated a slew of inappropriate comments, including misogynistic, racist, and anti-Semitic content, underscoring the gravity of the fairness challenge within AI deployment. Therefore, the principle of fairness is an irreplaceable role in the AI system and is vital to eliminate or alleviate the influence of biases that may cause harm to anyone related to the system. In the context of representation learning across domains, Fairness is widely applied to different domains so its definition may vary. However, it can be described as the impartial treatment of individuals or groups, without bias or favoritism, based on their inherent or acquired characteristics, especially in decision-making contexts [151, 186]. Particularly, fairness refers to the ability to correct data bias or algorithm bias,

especially in different domains. In other words, it aims to make the representations and model outputs invariant to sensitive attributes. As for more specific downstream tasks, like cross-domain face recognition, fairness have different goals, such as group fairness [44, 70], individual fairness [44], and counterfactual fairness [105], and some specific metrics [81, 258] for different tasks. For example, fairness through unawareness can effectively address group unfairness issues even in new domains, as long as the embeddings used do not contain inherent biased information, such as gender. Additionally, the Lipschitz condition can promote individual fairness in cross-domain areas by ensuring that similar individuals are treated similarly within different domains.

1.2.4 Explainability. With the widespread adoption of powerful deep learning models like CNN [107] and GAN [59], contemporary AI systems have become integral to our daily lives, aiding us in decision-making. However, the opaque nature of these complex deep learning models and the inherent black-box property raise concerns. These concerns, voiced by both industry and academia, highlight the lack of internal explanations in many AI systems. The inability to elucidate AI system behaviors has spurred the demand for transparency. Explainability emerges as a pivotal attribute in cultivating trustworthy AI systems. In the context of representation learning across diverse domains, the end-to-end nature of black-box deep learning models often renders learned representations less interpretable. This poses challenges in providing definitive evidence to understand model predictions. Consequently, the focus shifts toward enabling data representations to be more traceable. This traceability empowers researchers to gain deeper insights into learned data representations and offer more comprehensive explanations for model behaviors. For instance, techniques such as saliency maps [201] can highlight discriminative evidence within images across domains, bolstering the interpretability of models [297].

1.3 Related Surveys

Representation learning constitutes the foundational challenge in machine learning, and the pervasive integration of AI systems has harnessed the potency of machine learning to achieve notable performance levels, permeating various facets of our societal fabric. In pursuit of fostering trustworthiness, substantial efforts have been channeled, compelling us to address this very concern within representation learning. Particularly, when representation learning encounters cross-domain scenarios, its ability to generalize across varying domains often falters. This realization propelled us to introduce a comprehensive review framework for trustworthy representation learning across domains. While numerous surveys address trustworthy AI [95, 111, 135, 212], trustworthy graph neural networks learning [33, 264] and representation learning across domains [103, 224, 225, 276, 279], none have explored the specific domain we have introduced. Distinguishing itself from these surveys, our endeavor applies universal concepts within trustworthy AI and adapts them to the nuances of representation learning across diverse domains. By bridging this gap, our survey contributes a unique perspective to the intersection of these two crucial spheres.

1.4 Contributions

Our contributions are thus summarized as:

- An Instantiated Trustworthy Framework: We introduce the pioneering trustworthy representation learning across domains framework, outlining its application in various representation learning scenarios across domains. Guided by the principles of trustworthy AI, this framework serves as a roadmap, organizing directions and corresponding technologies in representation learning across domains.

Table 1. Illustration of the Typical Representation Learning Across Domains Scenarios

Scenarios	Training Data	Test Data	Condition
Domain Adaptation [225]	$\mathcal{D}_s, \mathcal{D}_t$	$\mathcal{D}_t$	$P(X_s, Y_s) \neq P(X_t, Y_s)$
Domain Generalization [224, 279]	$\mathcal{D}_{s_1}, \dots, \mathcal{D}_{s_n}$	$\mathcal{D}_t$	$P(X_{s_1}, Y_{s_1}) \neq \dots \neq P(X_{s_n}, Y_{s_n}) \neq P(X_t, Y_s)$
Cross-domain Few-shot Learning [67]	$\mathcal{D}_s, \mathcal{D}_t^l$	$\mathcal{D}_t^u$	$P(X_s, Y_s) \neq P(X_t^l, Y_t^l), \mathcal{Y}_s \cap \mathcal{Y}_t^l = \emptyset$

$\mathcal{D}_s$ denotes source domain and $\mathcal{D}_t$ indicates target domain. $\mathcal{D}_t^l$ represents few labeled target domain. In cross-domain few-shot learning scenario, $\mathcal{D}_t = \{\mathcal{D}_t^l, \mathcal{D}_t^u\}$, where $\mathcal{D}_t^l$ is few labeled target domain, which can be used in training stage, and $\mathcal{D}_t^u$ is unseen target domain used to evaluate the learned model.

- Comprehensive Review of Trustworthy Concepts: For each concept embedded within our trustworthy framework, we offer an exhaustive review, delving into the core concepts from fundamental definitions to intricate methodologies.
- Exploring Future Research Avenues: Through a holistic survey of the diverse representation learning across domains directions encapsulated by our trustworthy framework, we identify promising avenues for future research. By illuminating these potential trajectories, we aim to stimulate further exploration and advancements in the realm of trustworthy representation learning across domains.

The remainder of this survey is organised as follows. We first introduce the background of representation learning across domains and the concepts of trustworthy in our framework in Section 2. Then, Sections 3–6 summarize existing methods for trustworthy representation learning across domains from different concepts in trustworthy. We conclude the survey in Section 7.

2 BACKGROUND AND CONCEPTS

In this section, we will present the foundational background and essential concepts. Section 2.1 provides the essential notions related to trustworthy representation learning across domains. Sections 2.2 and 2.3 present problem definitions with representation learning across domains.

2.1 Notions and Definitions

We first give some notions and definitions that will be involved in this article. Let $\mathcal{X}$ be the input data space and $\mathcal{Y}$ be the label space. A domain can be defined as a joint distribution $P(X, Y)$ on $\mathcal{X} \times \mathcal{Y}$. For a specific domain $\mathcal{D} = \{X, Y\}$, we define the $P(X)$ as the marginal distribution on X , the $P(Y|X)$ as the conditional probability distribution of Y given X , and the $P(X|Y)$ as the class conditional distribution of X given Y . In general, we can learn the conditional probability distribution $P(Y|X)$ in a supervised way with a specified domain.

In the context of representation learning across domains, we assume to have two kinds of domains, i.e., source domain $\mathcal{D}_s$, which is usually full labeled, and target domain $\mathcal{D}_t$, which is usually unlabeled. Sometimes, the source domain may consist of more than one domain. In general, we hypothesize that there exists a domain shift between source and target domains, i.e., $P(X_s, Y_s) \neq P(X_t, Y_t)$. For source domain $\mathcal{D}_s$, it contains i.i.d. data-label pairs sampled from $P(X_s, Y_s)$, specifically, $\mathcal{D}_s = \{(x_s^i, y_s^i)\}_{i=1}^{n_s}$ with $(x_s^i, y_s^i) \sim P(X_s, Y_s)$. For target domain $\mathcal{D}_t = \{x_t^j\}_{j=1}^{n_t}$, the data are sampled from the marginal distribution $P(X_t)$ due to the label are inaccessible. In general, given the source domain $\mathcal{D}_s$ and target domain $\mathcal{D}_t$, the goal of representation learning across domains is to learn a model f with $\mathcal{D}_s$ or $\mathcal{D}_s$ and $\mathcal{D}_t$ such that the f can generalize well on the target domain $\mathcal{D}_t$.

Problems in representation learning across domains can be organized into several scenarios, such as domain adaptation [284], domain generalization [115], and cross-domain few-shot learning [67]. See Table 1 for an overview. We also consider their combinations with federated learning [165] and continual learning [110]. Different application scenarios of representation learn-

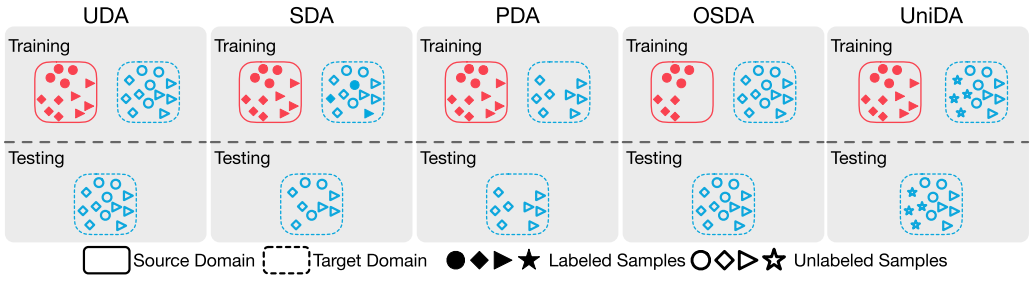


Fig. 2. Illustration of label space of unsupervised domain adaptation (UDA), Semi-supervised Domain Adaptation (SDA), Partial Domain Adaptation (PDA), Open-set Domain Adaptation (OSDA), and Universal Domain Adaptation (UniDA). Different colors represent different domains.

ing across domains contain different tasks. More details of these application scenarios and related tasks are given below:

2.2 Domain Adaptation

2.2.1 Existing Domain Adaptation Tasks. **Domain Adaptation (DA)** is the topic that has been broadly researched over the articles [52, 58, 112, 184, 194, 285, 288]. In general, given two domains, i.e., source domain $\mathcal{D}_s$, which is full labeled, and target domain $\mathcal{D}_t$, which is unlabeled or partially labeled. The goal is to develop a model that effectively harnesses data from both the source and target domains to achieve accurate classification of data in the target domain. The key hurdle in domain adaptation arises from the fact that data in these two domains are drawn from dissimilar distributions, compounded by the absence of label information from the target domain. This challenge necessitates innovative strategies to bridge this distribution gap and enable effective classification in the target domain. Learning the data presentations to alleviate the domain shift between two domains and adapting source knowledge to target domain is the major direction to solve the domain adaptation problem. Early, the main goal of the DA is to deal with the domain shift problem where $P(X_s, Y_s) \neq P(X_t, Y_t)$. Recently, DA [16, 89, 114, 163, 254] considers a more practical application situation that aims to address both domain shift problem and label shift problem, which means that the label spaces of source and target domains are not exactly same ($\mathcal{Y}_s \neq \mathcal{Y}_t$). Many existing DA methods fall under the umbrella of **Unsupervised Domain Adaptation (UDA)** [24, 43, 52, 58, 66, 109, 140, 184, 210]. These methods operate under the assumption that the label spaces of both domains are identical, aiming to mitigate the challenges posed by domain shift. However, **Partial Domain Adaptation (PDA)** [15, 16, 63, 112, 128, 238] relaxes this assumption by considering scenarios where the source domain's label space encompasses the target domain's label space. **Open-set Domain Adaptation (OSDA)** [10, 14, 89, 144, 162, 163, 230] addresses a more realistic application context where the label space of the target domain includes the label space of the source domain. In alignment with real-world needs, **Universal Domain Adaptation (UniDA)** [25, 50, 114, 182, 254, 286] relaxes label space constraints, accommodating private classes unique to each domain and shared common classes between them. These diverse methodologies cater to varying levels of label space alignment, catering to a wide spectrum of DA scenarios. Figure 2 illustrates the relationship between source and target label spaces in different DA tasks. We also summarize the difference among different tasks in Table 2.

2.2.2 Existing Domain Adaptation Methods. The ultimate goal lies in domain adaptation is to reduce the domain shift between source and target domains. To address this, Ben-David [12]

Table 2. Illustration of the Existing Domain Adaptation Tasks

Tasks	Label Set				Data Set		
	$Y_s = Y_t$	$Y_t \in Y_s$	$Y_s \in Y_t$	$Y_s \cap Y_t \neq \emptyset$	X_s full labeled	X_t full unlabeled	X_t few labeled
UDA [284]	✓	×	×	✓	✓	✓	×
SDA [112]	✓	×	×	✓	✓	×	✓
PDA [16]	×	✓	×	✓	✓	✓	×
OSDA [163]	×	×	✓	✓	✓	✓	×
UniDA [254]	×	×	×	✓	✓	✓	×

UDA means Unsupervised Domain Adaptation. **SDA** means Semi-supervised Domain Adaptation. **PDA** means Partial Domain Adaptation. **OSDA** means Open-set Domain Adaptation. **UniDA** means Universal Domain Adaptation. Y_s denotes source label space, and Y_t indicates target label space. X_s and X_t represent the source and target data, respectively.

proposed the bound for target risk $\epsilon^t(h)$ by using the available source risk $\epsilon^s(h)$ as follows:

$$\epsilon^t(h) \leq \epsilon^s(h) + d_{\mathcal{H}\Delta\mathcal{H}}(P(X_s), P(X_t)) + \lambda_{\mathcal{H}}, \quad (1)$$

where $\epsilon(h)^{s/t} := \epsilon(h, h^{s/t}) = \mathbb{E}_{x_{s/t} \sim P(X_{s/t})} [h(x_{s/t}) \neq h^*(x_{s/t})]$, the h can be any classifier from a hypothesis space $\mathcal{H}$ and the $h^{s/t}$ is the true classifier on the $P(X_{s/t})$. The $\mathcal{H}\Delta\mathcal{H}$ -divergence is defined as

$$d_{\mathcal{H}\Delta\mathcal{H}}(P(X_s), P(X_t)) := \sup_{h, h' \in \mathcal{H}} |\epsilon^s(h, h') - \epsilon^t(h, h')|, \quad (2)$$

and the $\lambda_{\mathcal{H}}$ denotes the difference in classifier across two domain and is written as

$$\lambda_{\mathcal{H}} := \inf_{h \in \mathcal{H}} |\epsilon^s(h) + \epsilon^t(h)|. \quad (3)$$

Inspired by this theoretical analysis, the existing DA methods mainly explore the direction of learning domain-invariant feature representations. In general, these methods can be roughly organized into two groups: moment-matching-based methods [139, 142, 218, 269] and adversarial learning-based methods [32, 52, 140, 145, 184, 217].

Moment-matching-based methods explicitly measure the discrepancy between the source and target domains on the learned data representations and match the discrepancy with well-defined moment-based distribution measurements [268]. The **Maximum Mean Discrepancy (MMD)** is one of the most typical measurements defined as

$$MMD(X_s, X_t) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(x_s^i) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(x_t^j) \right\|_{\mathcal{H}}, \quad (4)$$

where n_s and n_t are the number of data in the source and target domains, $\phi(\cdot)$ is the the kernel-induced feature map that projects from the **Reproducing Kernel Hilbert Space (RKHS)**, and $\|\cdot\|_{\mathcal{H}}$ is the RKHS norm. Early, Long et al. [139] proposed the **Deep Adaptation Network (DAN)** by designing the multiple kernel variant of maximum mean discrepancies [61] layers to project the hidden representations into an RKHS and match the mean embeddings of source and target domains. The authors of Residual Transfer Networks [141] further extended the DAN with a residual layer and relaxed the restriction on the classifier. Yan et al. [243] designed the Weighted Maximum Mean Discrepancy to address the class weight bias across domains problem by providing the class-specific auxiliary weights strategy to alleviate the drawback of MMD on class weight bias. Long et al. [142] considered to align source and target representations by optimizing the joint distributions of multiple domain-specific layers on a joint maximum mean discrepancy measurement with an adversarial training way.

Correlation Alignment (CORAL) is another measurement to align the second-order statistics (covariance) of source and target domains, which is defined as

$$CORAL(X_s, X_t) = \frac{1}{4d^2} \|C_s - C_t\|_F^2, \quad (5)$$

where d indicates the feature dimensionality, $\|\cdot\|_F^2$ represents the squared matrix Frobenius norm, and C_s and C_t denote the feature covariance matrices of X_s and X_t , respectively. The authors of Deep Correlation Alignment [210] revised the CORAL to nonlinear transformation by incorporating it to the last layer of the deep neural network and designing the corresponding differentiable loss function. Zhuo et al. [295] proposed the Deep Unsupervised Convolutional domain Adaptation, which extended the CORAL not only to fully connected layers in deep neural network but also to convolutional layers, and further improve the generalization of learned representations. Furthermore, Morerio et al. [154] jointly combined the correlation alignment and entropy minimization into a unified framework, i.e., Minimal-Entropy Correlation Alignment, with geodesic distances to align source and target domains in the learned representations. Chen et al. [20] presented the Joint Discriminative Domain Alignment framework, which consists of CORAL loss, instance-based discriminative loss, and center-based discriminative loss, to learn the discriminative feature representations across domains.

Beside the measurements of MMD and CORAL, there are many other measurements include but not limited to: (1) Zellinger et al. [260] proposed a new measurement, i.e., Central Moment Discrepancy, to match the high order central moments of probability distributions tried to match the higher-order moments by means of order-wise moment differences; (2) Chen et al. [21] investigated the higher order statistics, e.g., third- and fourth-order statistics, for domain alignment by designing the **Higher-order Moment-matching (HoMM)** framework and further extending the HoMM into RKHS; (3) Shen et al. [193] designed the Wasserstein Distance Guided Representation Learning framework to learning domain-invariant feature representations by minimizing the the estimated Wasserstein distance between source and target domains in an adversarial manner; (4) Jiang et al. [83] presented the Resource Efficient Domain Adaptation framework, which distills the transfer knowledge across source and target domains from the top layers to shallower layers in deep neural network by minimizing the Jensen-Shannon Divergence [51] between the probability distributions of student and teacher models. (5) Li et al. [125] induced the Adaptive Batch Normalization model to replace the statistics, e.g., moving average mean and variance, of Batch Normalization layers trained via source domain to these estimated from the target domain.

Adversarial learning-based methods are inspired by the GAN [59], which aim to reduce the domain shift across different domains via an adversarial manner. This mechanism has been proved to be the most influential mechanism over others in domain adaptation. Mathematically, the adversarial learning in domain adaptation can be formulated as a minimax optimization problem with two competitive loss objectives: (a) $\epsilon(C)$ on classifier C , which is optimized by minimizing the classification error on the source domain; (b) $\epsilon(D, G)$ on domain discriminator D and feature extractor G over source and target domains, which is minimized on D but maximized over G :

$$\epsilon(C) = \mathbb{E}_{(x_s^i, y_s^i) \sim \mathcal{D}_s} \mathcal{L}_{ce}(C(G(x_s^i), y_s^i)), \quad (6)$$

$$\epsilon(D, G) = -\mathbb{E}_{x_s^i \sim \mathcal{D}_s} \log[D(G(x_s^i))] - \mathbb{E}_{x_t^j \sim \mathcal{D}_t} \log[1 - D(G(x_t^j))], \quad (7)$$

where $\mathcal{L}_{ce}$ is the cross-entropy loss. The general formulation of minimax game in adversarial learning is defined as

$$\begin{aligned} & \min_{C, G} \epsilon(C) - \lambda \epsilon(D, G), \\ & \min_D \epsilon(D, G), \end{aligned} \quad (8)$$

where λ is the hyper-parameter to balance source risk and domain adversary. One of the earliest adversarial learning-based domain adaptation is Domain Adversarial Neural Network [53], which incorporated a minimax game framework into a single feed-forward network with a gradient reversal layer. While the other earliest method, i.e., Adversarial Discriminative Domain Adaptation [217], adopted an inverted label GAN loss to divide the optimization process into two independent steps for feature extractor and domain discriminator. Instead of only considering how to align the marginal distribution over source and target domains, Long et al. [140] jointly aligned the marginal distribution and conditional distribution with the proposed Conditional Domain Adversarial Networks, which includes a conditional discriminator with the concatenate input from feature representation and the corresponding classifier predictions. Zhang et al. [269] provided a new adversarial domain adaptation margin theory based on the hypothesis-induced discrepancy, and designed a novel measurement, i.e., Margin Disparity Discrepancy, to facilitate the minimax optimization with rigorous generalization bounds. Xu et al. [241] addressed two problems in adversarial domain adaptation, i.e., insufficient data from two domains to ensure domain-invariance over the whole representation space and hard label for the discriminator to judge real and fake data, and induced the adversarial Domain Adaptation with Domain Mixup framework, which includes a mixup version discriminator to guarantee domain-invariance in a more continuous latent space and result in a robustness model across domains. Wei et al. [232] alleviated the adverse effort of adversarial domain alignment induced by ignoring how to make the alignment proactively serve the classification task by proposing the Task-oriented Alignment to decompose the source feature and guide the alignment between source and target features with the prior knowledge from the classification task. Chen et al. [24] tackled the mode collapse problem caused by failing to make full use of predicted discriminative information and design a Discriminator-free Adversarial Learning Network which reuses the classifier to replace the discriminator and realizes domain alignment with discriminative feature presentations.

2.3 Domain Generalization

2.3.1 Existing Domain Generalization Tasks. **Domain Generalization (DG)** addresses a scenario where the target domain is not accessible during model training. In this setup, multiple fully labeled source domains $\mathcal{D}_{s_1}, \dots, \mathcal{D}_{s_{n_s}}$ are typically available. The key challenge in domain generalization revolves around devising a model that effectively generalizes to an unseen target domain $\mathcal{D}_t$ based on information from diverse source domains. DG is introduced as a solution to both the domain shift problem and the absence of target domain data during model training. Its significance has garnered significant interest from both academia and industry, owing to its potential for real-world applications [224, 279]. The initial focus of DG was on scenarios involving multiple source domains, each stemming from a distinct distribution. The core objective of DG entails training a model using these diverse source domains, with the aim of enabling it to generalize effectively to an unseen target domain. This specific DG variant is referred to as **Multi-source Domain Generalization (MSDG)** [4, 40, 64, 115, 155, 191, 275, 282, 283]. In MSDG, the central challenge revolves around training a model that can harness the information from multiple source domains to perform effectively on a target domain that has not been seen during training. Recently, a more challenging and realistic problem, i.e., **Single Domain Generalization (SDG)** [45, 80, 120, 171, 221, 231], is proposed for DG where only one source domain is accessible for model training. Meanwhile, some researchers further consider more challenging tasks that the label space of source and target domains are not consistent in MSDG and SDG, i.e., **Open-set Domain Generalization (OSDG)** [198] and **Open-set Single Domain Generalization (OSSDG)** [287]. In addition to this, the **Semi-supervised Multi-source Domain Generalization (SSMSDG)** [280] is proposed to address the task where the data in multi-source domain is

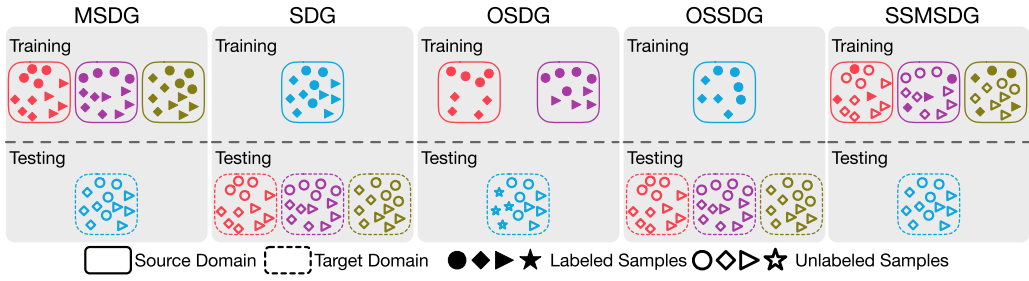


Fig. 3. Illustration of label space of Multi-source Domain Generalization (MSDG), Single Domain Generalization (SDG), Open-Set Domain Generalization (OSDG), Open-set Single Domain Generalization (OSSDG), and Semi-supervised Multi-source Domain Generalization (SSMSDG). Different colors represent different domains.

Table 3. Illustration of the Existing Domain Generalization Tasks

Tasks	Label Set				Data Set	
	$Y_s = Y_t$	$Y_t \in Y_s$	$Y_s \in Y_t$	$Y_s \cap Y_t \neq \emptyset$	X_s full labeled	X_s few labeled
MSDG [284]	✓	×	×	✓	✓	×
SDG [112]	✓	×	×	✓	✓	×
OSDG [16]	×	×	×	✓	✓	×
OSSDG [163]	×	×	✓	✓	✓	×
SSMSDG [254]	✓	×	×	✓	×	✓

MSDG means Multi-source Domain Generalization. SDG means Single Domain Generalization. OSDG Open-set Domain Generalization. OSSDG means Open-set Single Domain Generalization. SSMSDG means Semi-supervised Multi-source Domain Generalization. Y_s denotes source label space, and Y_t indicates target label space. X_s and X_t represent the source and target data, respectively.

few labeled. Figure 3 illustrates the relationship between source and target label spaces in different DG tasks. We also summarize the difference among different tasks in Table 3.

2.3.2 Existing Domain Generalization Methods. The primary objective of domain generalization is to train a model using either a single or multiple distinct but interconnected source domains in a way that it demonstrates strong generalization performance on an unseen target domain. The absence of target domain data during model training complicates the estimation of domain shift. In light of this challenge, DG centers on acquiring domain-invariant representations that preserve discriminative qualities on target data. Existing DG methods can be broadly categorized into several groups, including but not limited to feature alignment-based approaches, meta-learning-based techniques, and data augmentation-based strategies. These methodologies collectively address the task of enabling models to generalize effectively across domains, contributing to the broader landscape of domain generalization research.

Feature alignment-based methods aim to align features between different source domains, and hope the learned representations can be invariant to unseen target domain. Along this direction, there are a plethora of methods that adopt different domain alignment strategies, such as adversarial-learning-based and moment-matching-based, to reduce the domain shift among different source domains. This direction shares the similar motivation to domain adaptation, in some degree, the domain alignment strategies applied in domain adaptation can be used to domain generalization. For example, Li et al. [115] utilized the MMD to minimize the domain shift between source domains, and align the feature distribution to a prior distribution in an adversarial learning

manner. Li et al. [124] designed Conditional Invariant Deep Domain Generalization framework to learn domain-invariant representations via a conditional invariant adversarial network. Zhao et al. [275] jointly minimized the KL divergence between the conditional distribution of multiple source domains and applied adversarial learning to match the feature distributions in multiple source domains. Matsuura et al. [148] considered a practical application scenario where domain label information is unavailable, and proposed a framework that can jointly cluster source data into different latent domains and align them under the adversarial learning way. See References [224, 279] for comprehensive surveys on domain generalization)

Meta-learning-based methods have drawn increasing attention from the domain generalization community in recent years. The meta-learning is commonly known as learning-to-learn, which attempts to benefit the learning algorithm itself via multiple learning episodes by dividing multiple source data into meta-train $\mathcal{D}_{train}$ and meta-test $\mathcal{D}_{test}$ parts to simulate domain shift. Here, we loosely define the model parameters as θ , the overall meta-learning framework can be written as

$$\begin{aligned}\hat{\theta} &= \theta^* - \alpha \nabla_{\theta^*} \mathcal{L}_{outer}(\mathcal{D}_{test}; \theta^*), \\ \theta^* &= \theta - \beta \nabla_{\theta} \mathcal{L}_{inner}(\mathcal{D}_{train}; \theta),\end{aligned}\tag{9}$$

where α and β are hyper-parameters, $\mathcal{L}_{inner}(\cdot)$ and $\mathcal{L}_{outer}(\cdot)$ are designed different meta-learning objectives, e.g., cross-entropy objective. The $\mathcal{L}_{outer}(\cdot)$ aims to simulate domain shift and back-propagate the gradients all the way back to the original parameters such that model learned by $\mathcal{L}_{inner}$ can be improved. This learning process is a bi-level optimization and can be formulated as

$$\hat{\theta} = \theta - \alpha \frac{\partial (\beta \mathcal{L}_{inner}(\mathcal{D}_{train}; \theta) + \mathcal{L}_{outer}(\mathcal{D}_{test}; \theta^*))}{\partial \theta}.\tag{10}$$

One of the most popular meta-learning frameworks is proposed by Finn [48], i.e., **Model-agnostic Meta-learning (MAML)**, which separates data into meta-train and meta-test sets, and optimizes the model under meta-train and meta-test tasks to improve the generalization of the learned model. Inspired by this, Li et al. [113] are the first to apply MAML to domain generalization problems. They proposed the framework, i.e., **Meta-learning for Domain Generalization (MLDG)**, which divides multiple source domains into uncorrelated meta-train and meta-test to mimic the domain shift. The optimization goal is to learn a model with meta-train data that can get low error on meta-test data. In this manner, the learned model can generalize to unseen target domains. Balaji et al. [11] proposed a regularization-based meta-learning method for the classifier, which is different from MLDG meta-learned the entire model. Dou et al. [40] combine two complementary losses, which aim to regularize the semantic structure of the feature space, with MAML framework in such a way that the learned representations would keep semantic information across domains and generalize to unseen target domain. Du et al. [41] designed Meta Variational Information Bottleneck to jointly learn domain-invariant representation and improve the model performance by applying KL Divergence regularization between distributions of latent coding of the samples with the same category but from different domains. Du et al. [42] designed an extended version of batch normalization named MetaNorm. They utilize the meat-learning framework to learn adaptive statistics for batch normalization, which can improve the reliability and generalizability of learned representations.

Data augmentation-based methods usually combine the above-mentioned DG methods, such as feature alignment-based and meta-learning-based methods, to investigate the benefit brought by data augmentation [197]. The ultimate goal of data augmentation is to avoid overfitting problem and improve the generalizability of the learned model by generating plenty of diverse data to enrich the original data. Specifically, given the original data x , the data augmentation provides the

transformation function $Aug(\cdot)$ to generate new data $Aug(x)$ where x and $Aug(x)$ share same semantic information. Early data augmentation techniques focus on basic image manipulations, such as random flip, color space change or transformation, random crop, random rotation, noise injection, kernel filter, random erase, and so on. They can design different $Aug(\cdot)$ by including single or multiple image manipulations. But these $Aug(\cdot)$ are only suitable for certain scenarios. For example, rotation and flip are not applicable for digit classification problems, such as “6” and “9” images are easily confused after augmentation. Later, inspired by adversarial attacks [19], adversarial data augmentation [120, 220, 274] is proposed, which generates new data by adding sign-flipped gradients back-propagated from classifier to original data. The augmentation process is an iterative way,

$$x_{t+1} \leftarrow x_t + \eta \nabla_{x_t} \mathcal{L}(\theta; x_t), \quad (11)$$

where x_{t+1} denotes the augmented data with one iteration from x_t . A small number of iterations are needed to produce sufficient perturbations. θ denotes the parameters in model, and the $\mathcal{L}(\theta; x_t)$ is defined as

$$\mathcal{L}(\theta; x) = \underbrace{\mathcal{L}_{task}(\theta; x)}_{\text{Task}} - \alpha \underbrace{\mathcal{L}_{res}(\theta; f)}_{\text{Restrict}}, \quad (12)$$

where $\mathcal{L}_{task}$ is the task loss defined to optimize the model, e.g., cross-entropy loss, α is the hyper-parameter, f is the feature of original data x extracted by model. The second item in Equation (12) is noted as $\mathcal{L}_{res}(\theta; f) = \|f - f^*\|_2^2$, where f^* is the feature of augmented data x^* , which is generated via Equation (11) with original data x as input. This augmentation process can enrich the diversity of the original data while keeping the semantic information. However, these adversarial data augmentation methods fail to simulate the real-world domain shift, because the augmented data are visually similar to the original data, e.g., share the same image style. Meanwhile, another direction of data augmentation is proposed [244, 257, 281, 282], which considers to enrich the diversity of source domains by synthesizes new domains out-of-source distribution. For example, Zhou et al. [281] proposed the Deep Domain-adversarial Image Generalization to map the source domains to unseen domains by learning a transformation network in an adversarial manner. Yang et al. [244] design the Adversarial Teacher-Student Representation Learning to alternatively learn generalizable representations and generate novel-domain data. Recently, feature-level augmentation has been a fast growing area, motivated by mixup [263], which linearly interpolates any two data and their labels, respectively, to generate new data, Shu et al. [198] propose Domain-Augmented Meta-learning framework to learn generalizable representation by enrich source domains on feature-level via a new designed Dirichlet mixup and label-level though KL divergence. Meanwhile, Zhou et al. [283] mix the feature statistics, i.e., mean and standard deviation, of two instances with random ratio to implicitly synthesize new domains.

2.4 Others

Cross-domain Few-shot Learning. Very recently, many efforts [26, 67] have proven that most of existing few-shot learning [204, 229] methods fail to exhibit generalization capacity when there is a big domain shift between the source and target domains. To investigate this challenge, **Cross-domain Few-shot Learning (CDFSL)** [67, 215] has been proposed. In CDFSL, there are two typical stages involved: (1) meta-learning stage, which utilizes existing source domain $\mathcal{D}_s$ consists of plenty of base category classes to learn a model; (2) meta-testing stage, which adapts the learned model with few-labeled target domain $\mathcal{D}_t^l$ contains a set of novel classes, and hope this adapted model can generalize well to target domain. There is no overlap between the base category classes and the novel classes, i.e., $Y_s \cap Y_t = \emptyset$, the size of few-labeled target domain is much less than the size of the source domain, and the source and target domains have significant domain shift,

Table 4. Commonly Used Benchmark Datasets for Domain Adaption and Domain Generalization

Dataset	Domain	Class	Sample	Description
Office-Caltech [170]	4	10	2,533	Caltech, Amazon, Webcam, DSLR
Office-31 [170]	3	31	4,110	Amazon, Webcam, DSLR
PACS [172]	4	7	9,991	Art, Cartoon, Photos, Sketches
VLCS [97]	4	5	10,729	Caltech101, LabelMe, SUN09, VOC2007
Office-Home [79]	4	65	15,588	Art, Clipart, Product, Real
Terra Incognita [195]	4	10	24,788	Wild animal images at locations L100, L38, L43, L46
Rotated MNIST [231]	6	10	70,000	Digits rotated from 0° to 90° with an interval of 15°
DomainNet [175]	6	345	586,575	Infograph, Clipart, Painting, Quickdraw, Sketch, Real

e.g., source domain data come from natural images while target domain data come from medical images.

Federated Learning. Federated learning [149, 262] aims to sustain a robust global model by communicating with distributed clients who do not share their local data. A common challenge in federated learning is the non-IID data distribution, where the data distributions learned by different clients vary. Formally, A typical Federated Learning paradigm can be formalized as follows: Assume we have a set $\mathcal{C} = \{C_1, C_2, \dots, C_n\}$ of n different clients, where each client C_k has its private data $\mathcal{D}_k$ with the corresponding task $\mathcal{T}_k$, and a global model parameterized by θ_g deployed on a centralized server, which communicates with the clients during their training on their own local data. FL learns a global model that minimizes its risk on each of the client tasks,

$$\min_{\theta_g} \mathbb{E}_{\mathcal{T}_k \in \mathcal{T}} [\mathcal{L}_k(\theta_g)], \quad (13)$$

where $\mathcal{L}_k$ is the k th client objective $\mathcal{T}_k$.

2.5 Datasets and Open-source Implementation

In this section, we summarize several commonly used benchmark datasets in domain adaptation and domain generalization. In Table 4, we first provide an overview of some widely used detests of the image classification task, including Office-Caltech [170], Office-31 [170], PACS [172], VLCS [97], Office-Home [79], Terra Incognita [195], Rotated MNIST [231], and DomainNet [175]. Each dataset consists of images from n domains and data from different domains share the same label space. We also have categorized existing representative trustworthy representation learning methods across domains with available open-source implementations. For more details, please refer to Table 5.

3 ROBUSTNESS IN REPRESENTATION LEARNING ACROSS DOMAINS

The robustness is the fundamental element in the trustworthy AI framework. In general, it refers to capability of the AI system that remains high performance under different scenarios. It is vital for the AI system to establish well a reputation of trustworthiness in people and society in real-world applications. Ignoring the importance of robustness in AI system will raise massive problems when these systems are deployed to tasks related to safety [71], ethnicity [179], health [47], and so forth. As the results, the earned well reputation will be gradually eroded. In the field of representation learning across domains, the term robustness is to describe the capability of learned representations that lead to a robust model under different circumstances. For example, a self-driving model learned by using urban data should perform well under rural circumstance. A face recognition model trained using Asian face data should accurately recognize non-Asian faces.

In this section, we present a comprehensive review of robustness with domain adaptation, domain generalization, and others, from different aspects, such as pseudo-labeling, self-supervision,

Table 5. Categorization of Representative Trustworthy Representation Learning Methods Across Domains with Available Open-source Implementations

Trustworthy Concepts	Across Domains Scenarios	Representative Methods	Available Code
Robustness	Domain Adaptation	MSTN [239] DANCE [181] UAN [254] RFA [8]	Github Github Github Github
	Domain Generalization	ADA [220] M-ADA [171] MEADA [274] DAML [198]	Github Github Github Github
Privacy	Domain Adaptation	D-MCD [29] SHOT [127] SALAD [102] AaD [246]	Github Github Github Github
	Domain Generalization	SFDG [158] DEKAN [49] PromptStyler [28]	Github Github Github
Fairness	Domain Adaptation	FTS [27] FCR [6] FVM [202] RFCS [177]	Github Github Github Github
	Domain Generalization	EF [146] FATDM [167]	Github Github
Explainability	Domain Adaptation	SFIT [75]	Github
	Domain Generalization	EDCM [297]	Github

open-set, and so on. Instead of providing an exhaustive review on every direction, we aim to introduce the mainly focusing lines of the researches and try to lay out how the robustness problem in trustworthy representation learning across domains has been addressed in recent years. We also demonstrate some potential direction in robustness for future research.

3.1 Domain Adaptation

Even though typical DA methods have obtained significant achievement in many applications, they still need to improve their robustness in real-world application scenarios. In this section, we mainly focus on three branches to improve the robustness of DA: pseudo-labeling [22, 37, 84, 134, 183, 239, 267], self-supervision [96, 164, 181, 211, 286], and open-set [14, 25, 46, 50, 114, 133, 181, 182, 185, 254, 286]. Most of these methods combine with the typical DA methods, i.e., moment-matching-based and adversarial-learning-based, to jointly explore their advantage for robust domain adaptation. Recently, pseudo-labeling has been successfully applied to DA and gains promising performance on many DA tasks. The main challenge for pseudo-labeling-based methods is how to fully take advantage of pseudo-labeled target data to benefit the domain alignment instead of misleading domain alignment. Meanwhile, in more real-world situations where the labels in the target domain are not exactly the same as those in the source domains, researchers have come up with open-set related domain adaptation problems. These are designed to make existing

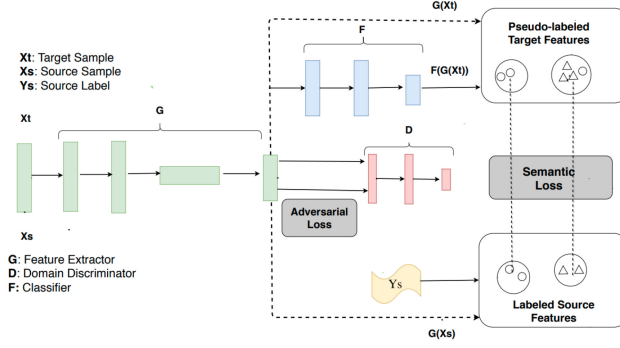


Fig. 4. Framework of Moving Semantic Transfer Networks (MSTN). (Image from Reference [239].)

adaptation methods stronger and more adaptable, considering differences in label spaces between different domains. This helps these methods work more robust in practical scenarios. The main goal is to accurately classify target data from the common label space shared with source label space while identify target data from the private label space different from source label space. Very recently, self-supervised learning has shown significant improvement on representation learning by fabricating pretext tasks to excavate supervisory signals from data itself instead of category information. Several endeavors have utilized self-supervised learning to DA problems and achieved significantly performance.

3.1.1 Robust Domain Adaptation with Pseudo-labeling. **Pseudo-labeling in DA** operates under the idea that certain target data can be given pseudo-labels and used for training. Saito et al. introduced an approach [183] involving two networks to create pseudo-labels for target data. Then, a third network is trained using these pseudo-labeled target data to develop representations that are specifically geared toward the target domain. Xie et al. [239] designed the **Moving Semantic Transfer Network (MSTN)** to realize class-level feature alignment between source and target distribution by assigning pseudo-labels to target data. See Figure 4 for an overview. Specifically, the MSTN jointly eliminates the domain shift between source and target domains via adversarial learning manner and align the centroids of source and target via semantic transfer objective. This objective is defined as

$$\mathcal{L}_{SM}(X_s, Y_s, X_t) = \sum_{k=1}^K \phi(C_s^k, C_t^k), \quad (14)$$

where K is the number of classes, $\phi(\cdot, \cdot)$ is any appropriate distance measure function mentioned in Section 2.2.2, and C_s^k and C_t^k denote the centroid of source and target for k th category, respectively. The target centroids are estimated by pseudo-labeled target data. To avoid the error accumulation problem and insufficient categorical information in each batch, they adopted exponential moving average strategy to estimate centroids for source and target domains, which is formulated as

$$\begin{aligned} C_s^k &\leftarrow \gamma C_s^k + (1 - \gamma) C_{s(batch)}^k, \\ C_t^k &\leftarrow \gamma C_t^k + (1 - \gamma) C_{t(batch)}^k, \end{aligned} \quad (15)$$

where γ is the hyperparameter controls the contribution of each batch to estimate centroids. $C_{s/t(batch)}^k$ is the mean value of data in source/target batch that belong to k th class. Later, how to safely assign pseudo-label to target data and select pseudo-labeled target data has been explored by many works. For example, Zhang et al. [267] designed the Incremental Collaborative and Adversarial Network to iteratively sample a subset of pseudo-labeled target data with a classification

or discriminator confidence scores-based criterion to retrain the model. Den et al. [37] proposed Cluster Alignment with a Teacher to investigate the class-wise alignment between source and target distribution by reliably assigning pseudo-labels to target data with a teacher classifier. To further alleviate the error accumulation induced by unreliable pseudo-labels on target data, Chen et al. [22] presented the Progressive Feature Alignment Network [22]. This approach involves a step-by-step process to choose trustworthy pseudo-labeled target data using cross-domain similarity measurements. It also dynamically adjusts the alignment of source and target class centers to enhance the alignment between different domains. Jiang et al. [84] addressed the error accumulation problem from sample selection aspect. Specifically, they presented a sampling-based implicit method, which utilizes pseudo-labels implicitly to guide data sampling and align the joint distribution between features and labels. Liu et al. [134] provided a cycle self-training-based method, which improves the reliability of target pseudo-labels by inducing the Tsalli entropy regularization, and promotes the generalizability of pseudo-labels across domains.

3.1.2 Robust Domain Adaptation with Self-supervision. **Self-supervision in DA** shares the spirit of the self-supervised learning [87], and incorporates the self-supervised-learning pretext tasks, e.g., image rotation [57], patch location prediction [159], and instance discrimination [236] (used for contrastive learning), into existing DA methods to improve their robustness. Early, Sun et al. [211] added auxiliary self-supervised tasks, such as rotation prediction, flip prediction, and patch location prediction, for source and target domains. See Figure 5 for details. In this framework, the over objective is written as

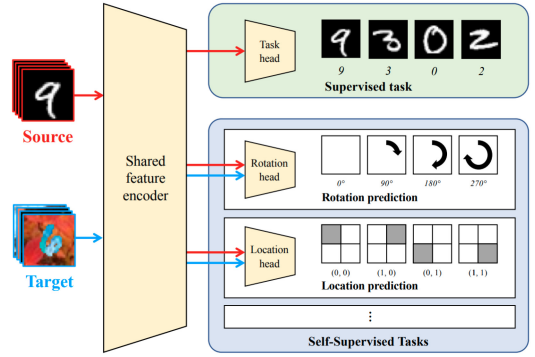


Fig. 5. Self-supervised-learning-based framework for unsupervised domain adaptation. (Image from Reference [211].)

$$\min_{\theta, h_k, k=0 \dots K} \mathcal{L}_0(X_s; \theta, h_0) + \sum_{k=1}^K \mathcal{L}_k(X_s, X_t; \theta, h_k), \quad (16)$$

where θ is parameters of feature encoder, h_k denotes the parameters of the k th head task model, e.g., rotation prediction and location prediction task models, h_0 is the actual prediction task with objective $\mathcal{L}_0$, e.g., cross-entropy objective, and $\mathcal{L}_k$ is the predefined loss function for k th head task model and feature encoder. By optimizing the objective, the model will encourage both domains to align together along the direction related to tasks. Recently, inspired by the success of contrastive learning in self-supervised learning, Park et al. [164] proposed the Joint Contrastive Learning to learn discriminative feature representations, which investigates the mutual information between features and labels under the contrastive learning framework.

Different from previous DA methods, Saito et al. [181] proposed a universally applicable domain adaptation framework, named as **Domain Adaptative Neighborhood Cluster via Entropy Optimization (DANCE)**, to tackle arbitrary category shift. See Figure 6 for an overview. The over-optimization function is defined as

$$\mathcal{L}_{all} = \mathcal{L}_{ce} + \lambda(\mathcal{L}_{nc} + \mathcal{L}_{es}), \quad (17)$$

where λ is the hyperparameter, $\mathcal{L}_{ce}$ is the cross-entropy loss function, $\mathcal{L}_{nc}$ denotes the proposed neighborhood clustering loss function (see Figure 6 left part for an overview), and $\mathcal{L}_{es}$ represents

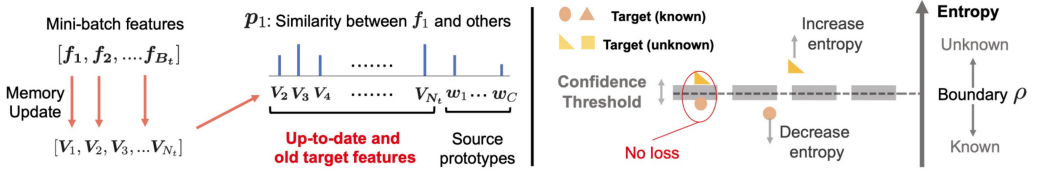


Fig. 6. Illustration of Domain Adaptive Neighborhood Clustering via Entropy Optimization (DANCE). **Left:** Similarity distribution calculation in neighborhood clustering. **Right:** Overview of the entropy separation loss. (Image from Reference [181].)

the well-designed entropy separation loss function (see the right-hand side of Figure 6 for an overview). The goal of $\mathcal{L}_{nc}$ is to learn discriminative and generalizable representation for target domain, which is defined as

$$\mathcal{L}_{nc} = -\frac{1}{|B_t|} \sum_{i \in B_t} \sum_{j=1, j \neq i}^{N_t+K} p_{i,j} \log(p_{i,j}), \quad (18)$$

where B_t is the sets of target samples' indices in the batch, N_t is the number of target data, K is the number of source classes, and $p_{i,j}$ denotes the probability of the target feature f_i is the neighbor of j th item in memory bank V , which stores all target features and prototype weight vectors. The $p_{i,j}$ is computed by

$$p_{i,j} = \frac{\exp(V_j f_i / \tau)}{\sum_{j=1, j \neq i}^{N_t+K} \exp(V_j f_i / \tau)}. \quad (19)$$

To separate target data into known classes and unknown classes, the entropy separation loss function $\mathcal{L}_{es}$ is formulated as

$$\mathcal{L}_{es} = \frac{1}{|B_t|} \sum_{i \in B_t} \mathcal{L}_{es}(p_i), \quad \mathcal{L}_{es}(p_i) = \begin{cases} -|H(p_i) - \rho|, & |H(p_i) - \rho| > m, \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

In $\mathcal{L}_{es}$, p_i is the output of classifier for target feature f_i , $H(p_i)$ is the entropy value of p_i , and ϕ is threshold to separate target data into known and unknown classes defined as $\frac{\log(K)}{2}$. The overall functions aim to learn the representations for target domain by incorporating contrastive learning to dig the similarity of target data and source class prototypes. Zhu et al. [286] further extended DANCE by providing different neighbors for different target data to excavate the similarity under the contrastive learning framework and improve the consistency of classifier for each target under different augmentations. Kim et al. [96] proposed Cross-Domain Self-supervision to learn discriminative and domain-invariant representations by jointly mining the similarity of source and target data in-domain and cross-domain under the contrastive learning.

3.1.3 Robust Domain Adaptation with Open-set. Open-set in DA aims to address the practical problem where source and target have different label spaces and promote the robustness of DA in real-world application. In general, open-set in DA considers two kinds of tasks: the first one is OSDA, which assumes that the target label space contains unknown classes for source label space. The later one is UniDA, which removes the restrictions on both label spaces, which allows source and target domains not only have their private label spaces but also share a common label space.

For OSDA, the early method [185] is based on adversarial learning framework, which aligns target data with source domain or reject them as the unknown target classes, which do not belong to source label space. See Figure 7 for an overview. They designed the classifier C to classify data

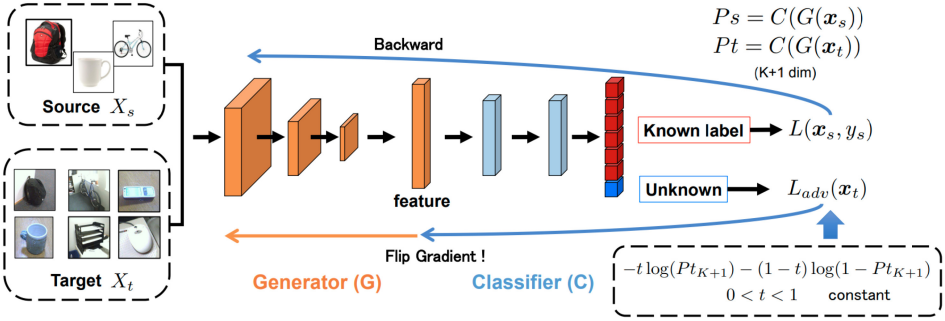


Fig. 7. Adversarial learning-based framework for open-set domain adaptation. (Image from Reference [185].)

into $K + 1$ classes where K is the number of classes in source label space while 1 indicates the target data from unknown classes. The overall objective function is defined as

$$\begin{aligned} \min_C \mathcal{L}_{ce}(x_s, y_s) + \mathcal{L}_{adv}(x_t), \\ \min_G \mathcal{L}_{ce}(x_s, y_s) - \mathcal{L}_{adv}(x_t), \end{aligned} \quad (21)$$

where $\mathcal{L}_{ce}(\cdot, \cdot)$ is the cross-entropy loss, and $\mathcal{L}_{adv}(\cdot)$ is a binary cross-entropy loss formulated as

$$\mathcal{L}_{adv}(x_t) = -\alpha \log(p(y = K + 1|x_t) - (1 - \alpha) \log(1 - p(y = K + 1|x_t))), \quad (22)$$

where $p(y = K + 1|x_t)$ denotes the probability of x_t being classified as the unknown class. α is designed to train the classifier to output $p(y = K + 1|x_t) = \alpha$, which is set to 0.5. Under the adversarial optimization process, the learned representations would gradually separate the target data into known and unknown classes. For these known classes data, they will be aligned to source data. Later, Feng et al. [46] and Liu et al. [133] shared the similar idea of separating target data into known and unknown classes, and aligning the target data from known classes to source data. Both of them are based on adversarial learning framework. In one approach, researchers delved into the semantic structure of both the source and target domains. They achieved this by aligning the centroid of target data from recognized classes with their corresponding centroid from the source domain. This strategy also involved pushing target data from unfamiliar classes away from the decision boundary. In another approach, a coarse-to-fine weighting mechanism was introduced. This technique gradually separated target data into known and unknown classes. Then, they devised a weighted adaptation process to align the distributions of target and source data, refining the alignment step by step. Recently, instead of using adversarial learning, Bucci et al. [14] proposed a self-supervision-based framework, which uses rotation recognition as the pretext task, to separate known and unknown target data and align the distributions of source and known target. To explore the useful information in target data belong to unseen classes, Jiang et al. [89] proposed a novel problem, termed as Semantic Recovery Open-Set Domain Adaptation, which is based on the OSDA problem definition and further provides the semantic attribute annotation on source domain. They designed a framework to simultaneously learn domain-invariant representation and build visual-semantic projection to restore the lacking of semantic attribute of target unknown classes.

For UniDA, a more practical and challenging problem than OSDA, the initial UniDA approach is **Universal Adaptation Network (UAN)** [254], which utilizes a metric comprising entropy and domain similarity values to assess the transferability of each target data and differentiate between target data belonging to common classes and those belonging to private classes. See Figure 8 for an overview. In the training phase, the overall optimization objective is defined in an adversarial

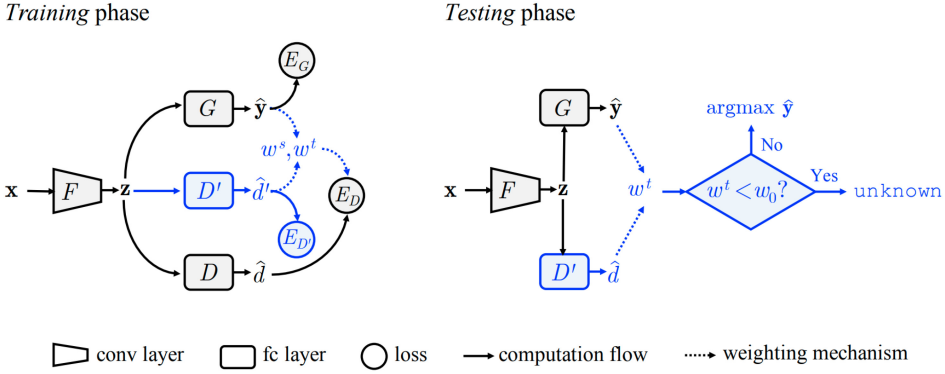


Fig. 8. Framework of Universal Adaptation Network (UAN). F is the feature extractor, G indicates the classifier, D represents the adversarial domain discriminator, and D' denotes the non-adversarial domain discriminator, which is used to compute the domain similarity value. (Image from Reference [254].)

manner:

$$\max_D \min_{F, G} E_G - \lambda E_D, \quad \min_{D'} E_{D'}, \quad (23)$$

where E_G , E_D , and $E_{D'}$ are the error for classifier G , adversarial domain discriminator D , and non-adversarial domain discriminator D' . They are defined as

$$\begin{aligned} E_G &= \mathbb{E}_{(x_s, y_s) \sim \mathcal{D}_s} \mathcal{L}_{ce}(G(F(x_s)), y_s), \\ E_D &= -\mathbb{E}_{x_s \sim \mathcal{D}_s} \omega_s(x_s) \log D(F(x_s)) - \mathbb{E}_{x_t \sim \mathcal{D}_t} \omega_t(x_t) \log (1 - D(F(x_t))), \\ E_{D'} &= -\mathbb{E}_{x_s \sim \mathcal{D}_s} \log D'(F(x_s)) - \mathbb{E}_{x_t \sim \mathcal{D}_t} \log (1 - D'(F(x_t))), \end{aligned} \quad (24)$$

where $\omega_s(x_s)$ and $\omega_t(x_t)$ are two sample-level transferability criterions for source and target data, which are used to constrain the D to distinguish the source and target data in the common label space. They are defined as

$$\begin{aligned} \omega_s(x_s) &= \frac{H(G(F(x_s)))}{\log(C_s)} - D'(F(x_s)), \\ \omega_t(x_t) &= D'(F(x_t)) - \frac{H(G(F(x_t)))}{\log(C_s)}, \end{aligned} \quad (25)$$

where $H(\cdot)$ is the entropy value and C_s is the number of classes in source domain. By utilizing these two criterions, UAN can disentangle source and target data into common classes and their private classes. In this way, the category gap and domain shift are jointly reduced. However, the proposed metric is less robust when entropy value is less reliable and domain similarity value is overconfident. Fu et al. [50] proposed **Calibrate Multiple Uncertainties (CMU)** to offer a more robust metric, which consists of entropy, confidence, and consistency from an ensemble framework. Different from UAN and CMU, Saito et al. [181] deal with the UniDA problem by utilizing contrastive learning to learn generalizable and discriminative representations and using entropy value of classifier to separate target data into known and unknown classes. Inspired by Reference [181], Zhu et al. [286] proposed a more robust framework to learn better representations and improve the robustness of entropy value of classifier by inducing consistency regularization on target data under different data augmentation. Recently, Li et al. [114] investigate discriminative clusters in source and target domains to discover the intrinsic structure of target domain and separate target data into known classes and unknown classes by exploiting domain consensus knowledge.

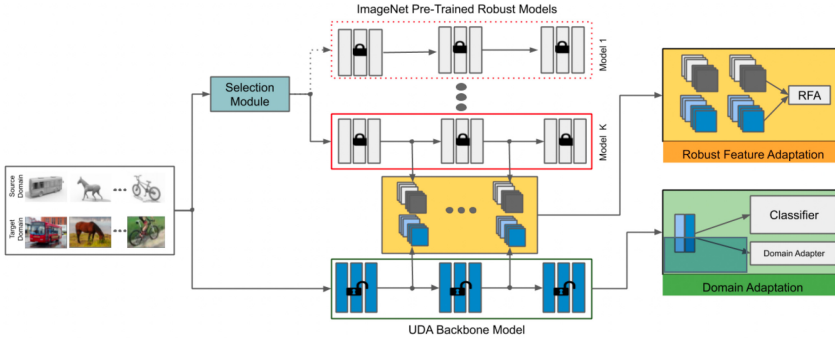


Fig. 9. Framework of Robust Feature Adaptation (RFA). (Image from Reference [8].)

Saito et al. [182] proposed One-vs-All Network to adaptively learn the threshold to separate target data into known and unknown classes by exploring the inter-class distance between source categories. However above methods ignore the importance of utilizing the intrinsic manifold structure relationship between source and target domains, which could lead their model less robust to real-world application. To address this issue, Chen et al. [25] proposed a global joint local learning framework, which includes geometric adversarial learning for domain alignment, subgraph-level contrastive learning for local region aggregation, and universal incremental classifier for automatic unknown threshold learning, to better align target data from known classes to source data while separate these from target unknown classes data.

3.1.4 Robust Domain Adaptation with Adversary Attacks. **Adversary attacks in DA** aims to improve the adversarial robustness of the domain adaptation models, which is crucial in real-world applications. Recent studies imply that domain adaptation methods are vulnerable to adversarial examples, i.e., inputs with a slight but intentional perturbation are incorrectly classified by the learned model. Such vulnerability make it risky for some security-related applications, e.g., face presentation attack detection [222] and semantic segmentation in autonomous cars [245]. Mehra et al. [150] investigated the impact of adversarial attacks on well-established UDA methods, employing benchmark datasets where these methods have previously demonstrated success. Zhang et al. [266] conducted the comprehensive study on the robustness of domain adaptation methods to adversarial attacks. They assessed representative domain adaptation approaches under different adversarial attack scenarios, including white-box [17] and black-box [153] attacks. Additionally, they introduced a cross-domain attack method leveraging pseudo labels to enhance the robustness of domain adaptation methods against adversarial samples. Later, instead of using a supervised loss function for adversarial sample generation, Awais et al. [8] improve Unsupervised Domain Adaptation robustness by aligning UDA method features with robust ImageNet model representations. See Figure 9 for an overview. The robust feature adaptation (RFA) loss of l th layer is defined as

$$\mathcal{L}_{RFA} = \frac{1}{b^2} \|G_T^l - G_S^l\|_F^2, \quad (26)$$

where $\|\cdot\|_F$ is the Frobenius norm, b is the batch size, and G_T^l and G_S^l represent the normalized activations of l th layer from ImageNet pre-trained robust models and the UDA backbone model, respectively. The aim of $\mathcal{L}_{RFA}$ is to fine-tune the UDA feature extractor to incorporate both robust features from adversarially trained models and domain-invariant features from UDA training. Lo et al. [138] introduced a method called Adversarially Robust Training for Unsupervised Domain Adaptation by enhancing robustness of the the UDA models via adversarial training for unlabeled

data. Agarwal et al. [2] further explored robustness of model in Unsupervised Domain Adaptation without Source Data. Their finds include the effective transfer of robust source models to the target and the significant benefits of using non-robust pseudo-labels and pair-wise contrastive loss in the adaptation process.

3.2 Domain Generalization

Compared with domain adaptation, DG is more difficult to learn a robust model that can generalize to target domain. Because it has to not only address the domain shift between source and target domains but also face the absence of target domain in training stage. Although plenty of existing DG methods have achieved significant performance, there is still a long way to realize robustness for domain generalization in real-world application. In this section, we mainly focus on two branches to review how to learn robust representations in DG under the real application scenarios: single-source-domain-based methods and open-set related methods. Different from the assumption in typical domain generation methods that these are multiple source domains can be used to learn the model, single-source-domain-based methods consider a more challenging and realistic problem, which assumes only one labeled source domains is available for training and many unseen target domains for test. Very recently, open-set related tasks have been proposed under the domain generalization, which require the learned model can accurately classify each target data into either a known class in source label space or an unknown class.

3.2.1 Robust Domain Generalization with Single-source Domain. **Single-source domain in DG** addresses a practical application scenario in DG that only one source domain is available to train the model. This problem is also well-known as SDG, which is the worst-case domain generalization scenario. One of the earliest SDG methods, named as **Meta-learning-based Adversarial Domain Augmentation (M-ADA)**, is proposed by Qiao et al. [171], which adopts the **adversarial data augmentation (ADA)** [220] into meta-learning framework and utilizes an auxiliary Wasserstein autoencoder to help ADA generate more challenging data to enrich the diversity of source domain. See Figure 10 for details. They relaxed the ADA objective in Equation (5) to

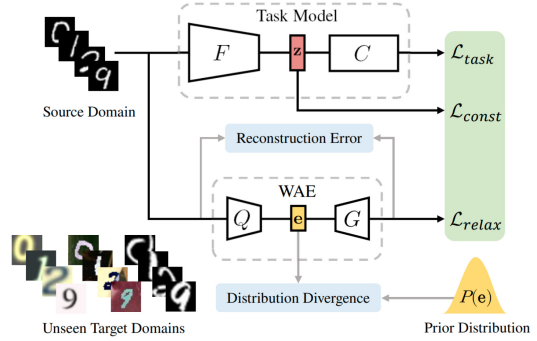


Fig. 10. Framework of Meta-Learning-based Adversarial Domain Augmentation (M-ADA). F is the feature extractor, C represents the classifier, and Q and G denote encoder and decoder, respectively. (Image from Reference [171].)

$$\mathcal{L}(\theta; x) = \underbrace{\mathcal{L}_{task}(\theta; x)}_{\text{Task}} - \underbrace{\alpha \mathcal{L}_{res}(\theta; f)}_{\text{Restrict}} + \underbrace{\beta \mathcal{L}_{rel}(\psi; x)}_{\text{Relax}}, \quad (27)$$

where ψ denotes the parameters in encoder Q and decoder G . $\mathcal{L}_{rel}$ ensures large domain transportation defined as

$$\mathcal{L}_{rel}(\psi; x) = \|x - G(Q(x))\|, \quad (28)$$

here Q and G are pre-trained to capture the distribution of the source domain. Adopting the same augmentation process as ADA in Equation (11), the M-ADA will generate diverse data to enrich

source distribution. They further jointed source data and its augmented version into a meta learning framework to learn the model that can generalize to unseen target domain. Zhao et al. [274] proposed **Maximum-entropy Adversarial Data Augmentation (MEADA)** to generate effective new data distributions consisting of “hard” data that are significantly different from source data by adding the maximum-entropy regularizer to ADA. Fan et al. [45] explored the impact of the statistics, i.e., standardization and rescaling statistics, of normalization layers to improve the generalization of the model. They proposed the Adaptive Standardization and Rescaling Normalization to complete the ADA for data augmentation by learning related statistics in network. Wang et al. [231] proposed a style-complement module to improve the capability of model on generating new data that are out of source distribution. In their framework, a mutual information-based min-max game is played to enhance the ability of style-complement module on data generation and promote the generalizability of model on unseen target domain. Li et al. [120] designed Progressive Domain Expansion Network to jointly generate new domains in a progressive manner via the domain expansion subnetwork and learn domain-invariant representation though a contrastive learning-based subnetwork. Cugu et al. [80] apply a diverse set of visual corruptions to enrich source distribution and propose an attention consistency loss to align the class activation maps between original data and the corrupted version. Different from above data augmentation-based methods, Wan et al. [221] address the SDG problem from the aspect of model architecture, they proposed a new convolutional model to extract unbiased universal feature for unseen target data.

3.2.2 Robust Domain Generalization with Open-set. **Open-set in DG** simulates the real-world application scenarios that unseen target domain has domain shift and a different label space from the source domains. This practical problem is difficult for existing typical DG methods. Because their learned domain-invariant representations can generalize to target domain is based on the assumption that source and target domains share the same label space. When target domain contains data not belong to source label space, the learned representation would likely fail to generalize to target domain. Very recently, there are two works [198, 287] try to tackle the open-set problem from different aspects. Shu et al. [198] address the open-set problem under the assumption that there are multiple source domains available for model training. They proposed **Domain-augmented Meta-learning (DAML)** to learn generalizable representation by enriching source domains on feature-level via a new designed Dirichlet mixup and label-level though KL divergence. While, Zhu et al. [287] tackle the open-set problem under the assumption that there only one source domain accessible for model training. They proposed CrossMatch framework, which jointly generates auxiliary data to simulate target unknown classes data and applies consistency regularization on generated auxiliary data between multi-binary classifiers and the model to improve the capability of learned model on unknown class identification. See Figure 11 for an overview. One of their major contribution is to generate auxiliary data that potentially out of source distribution to simulate unknown classes data. They followed the same strategy in adversarial data augmentation [220] to generate auxiliary data. Similar to Equation (12), their augmentation process is written as

$$x_u^{t+1} \leftarrow x_u^t + \eta \nabla_{x_u^t} \{ \mathcal{L}_{unk}(x_u^t) + \gamma \mathcal{L}_{res}(\theta_g; x_u^t, x_s) \}, \quad (29)$$

where $\mathcal{L}_{res}$ is the worst-case guarantee defined in Equation (12). $\mathcal{L}_{unk}$ is their designed auxiliary data generalization objective, which is formulated as

$$\mathcal{L}_{unk}(x_s, y_s) = -\log(p_{b,s}^{y_s}(t = 0|x_s)) + \sum_{i \neq y_s}^k \log p_{b,s}^i(t = 1|x_s), \quad (30)$$

where $p_{b,s}^{y_s}(t = 0|x_s)$ and $p_{b,s}^{y_s}(t = 1|x_s)$ represent the probabilities of being y_s th class and other classes from the y_s th binary classifier, respectively. By maximizing $\mathcal{L}_{unk}$ at the maximization stage,

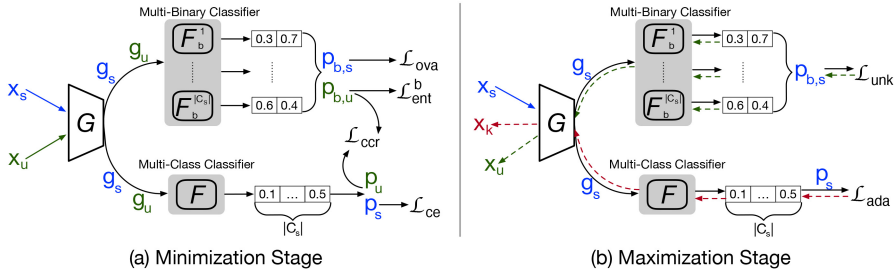


Fig. 11. Framework of CrossMatch. G is the feature extractor, F represents the classifier, multi-binary classifier is denoted as $F_b = \{F_b^1, \dots, F_b^{|C_s|}\}$ where $|C_s|$ is the number of classes in source label space. (Image from Reference [287].)

it encourages the model to generate new samples that all binary classifiers in F_b predict it as “other class.” Thus, the new generated sample could belong to unknown classes.

3.3 Others

Apart from the above-mentioned directions in representation learning across domains, which aim to improve the robustness in representation learning in real-world application scenarios, we also provide two new directions: cross-domain few-shot learning and federated learning across domain.

The CDFSL [26, 35, 67, 76, 123, 126, 215, 223] aim to understand and address the domain shift problem for few-shot learning [204, 229]. Chen et al. [26] found that existing few-shot learning methods fail to address the domain shift problem. Guo et al. [67] proposed the Broader Study of Cross-domain Few-shot Learning benchmark, which included image data from diverse domains and evaluated several methods, e.g., meta-learning methods, transfer learning methods, few-shot learning method, and new CDFSL methods. Tseng et al. [215] utilized feature-wise transformation layers to generate various feature distribution to improve the generalization capability of the learned model. Inspired by adversarial data augmentation [220], Wang et al. [223] proposed an adversarial meta task augmentation-based framework to alleviate inductive bias problem and improve the robustness of the model under domain shift. Liang et al. [126] proposed a Novel-enhanced Supervised Autoencoder to help model capture broader variations of the feature representation and increase the discrimination and generalization capability of learned model on target domain. Recently, Das et al. [35] used contrastive learning to learn a generalizable representation and train a masking module to select features that can benefit target domain classification.

The Federated Learning Across Domains [137, 165, 249, 265] aims to address the domain adaptation and domain generalization problems under the data privacy constraint. Although federated learning [262] has addressed data privacy and efficiency problems under the networks of distributed devices for deep learning. The learned model still suffers from domain shift problem, which would degrade the robustness of the model in real-world application scenarios. In the context of federated learning, addressing domain adaptation challenges, Pent [165] introduced a novel concept known as Unsupervised Federated Domain Adaptation. This approach involves transferring knowledge from decentralized nodes to a new node with a distinct data domain, referred to as the target domain. To tackle domain shift in federated learning, they introduced Federated Adversarial Domain Adaptation, which employs a dynamic attention model to effectively manage the domain shift problem. Different from Reference [165], Zhang et al. [265] addressed the domain generalization problem under federated learning. They proposed FedADG framework, which including **Adversarial Learning Network (ALN)** for each node to align each source domain to

the generated reference distribution, and a distribution generator in server to generated reference distribution.

3.4 Future Research Direction of Robustness

Novel Categories Recognition. In representation learning across domains, a key task within the open-set framework is identifying target data that lies outside the source label space. These endeavors [46, 50, 114, 185, 198, 287] typically center on mining valuable insights from data to aid in classifying target data into either known or unknown categories. In some cases, augmenting data with more information can facilitate the development of robust representations, pertinent to real-world applications. For instance, Jiang et al. [89] introduced semantic attribute annotations for the source domain. This approach yielded a more resilient model compared to existing methods. Another avenue for inducing meaningful information involves leveraging vision and language models [82, 118, 174]. These models are trained using image-caption pairs, offering beneficial impacts on various computer vision tasks, including object detection [277] and object classification [240]. This alternative direction has the potential to enrich the representations and enhance the performance of models in a range of applications.

Few Labeled Data. In the realm of representation learning across domains, a prevailing assumption is that target data remains unlabeled. However, fully annotating all target data can be both time-consuming and costly in practical applications. In many scenarios, having access to a small set of labeled target data is feasible and economical. Exploring effective strategies to maximize the utility of these limited labeled target data represents a promising avenue to enhance the robustness of learning representations. This direction holds potential for further advancements in representation learning across domains.

Feature Generation. The efficacy of data augmentation in bolstering the robustness of representation learning across diverse domains has been firmly established [198, 257, 274, 281]. These methods often focus on generating new data to enrich the diversity of the original data distribution. However, the concept of feature generation has received relatively limited attention. Only a handful of studies have explored this aspect, yielding substantial improvements. For instance, Xu and collaborators [240] introduced a model based on conditional variational autoencoders to generate new features that complement the original feature distribution. Li et al. [119], however, harnessed conditional Wasserstein generative adversarial networks to produce a variety of distinctive and discerning features. These efforts emphasize the untapped potential of enhancing features through data augmentation, offering a unique perspective on advancing representation learning across domains.

Foundation Models. Recently, foundation models such as **Contrastive Language-Image Pre-Training (CLIP)** [174] and **Segment Anything Model (SAM)** [98] have demonstrated remarkable capabilities in generalization, particularly in zero-shot and few-shot learning scenarios. However, these foundation models still face challenges in cross-domain learning tasks. Several studies [56, 199, 259] have highlighted that the generalization of these foundation models needs improvement, especially when dealing with tasks involving significant domain shifts between the source and target domains. Consequently, a promising avenue of research involves enhancing the generalization of foundation models for downstream tasks in the presence of domain shifts. This represents an emerging direction worth exploring.

4 PRIVACY IN REPRESENTATION LEARNING ACROSS DOMAINS

Deep neural networks are experiencing a surge in popularity across diverse domains including natural language processing, computer vision, rapid recognition, and recommendation systems. The wide use of machine learning models, at the same time, also brings privacy concerns. In recent

years, customers are gradually provided with access to software interfaces (e.g., from Google, Microsoft, Amazon) that allow them to easily use machine learning tasks for their own applications. However, if the training data are able to be recovered by some malicious users, serious issues can be caused by such information leaking, especially when the training data is sensitive, say, medical records. Likewise, the parameters of the model, which are considered as proprietary information, should also be protected from being stolen with access to the interface. In representation learning across different domains, due the information exchange among multiple domains, the risk of potential information leakage can be increased and requires more attention. Within this section, we initially provide a concise overview of common privacy challenges. Subsequently, our focus shifts toward addressing privacy concerns when undertaking cross-domain learning. Concluding, we shed light on less-explored challenges and potential avenues for forthcoming research.

4.1 Privacy in Deep Learning

In this section, we delve into the multifaceted landscape of privacy threats associated with deep learning models [36]. These threats are intricately categorized into three classes. The first class involves the inference of sensitive information about the underlying population. The second class pertains to the potential exposure of details within the training dataset, while the third class raises concerns about the disclosure of confidential model parameters.

For **Inference of the Population**, the exploration primarily delves into three key aspects: (1) *Statistical disclosure*: Given the trained model, the adversary tries to learn about the input of the model. Theoretically, to control statistical disclosure, it is required that applying the trained model should not reveal more knowledge about its training input than it is not applied. (2) *Model inversion*: The adversary seeks to deduce the sensitive attributes of the input to the model using the model's output, provided the input. (3) *Inference of class features*: Relaxing the sensitive attributes in model inversion to arbitrary class attributes, which characterizes a particular task, the adversary tries to learn about the characters of each class with access to the model.

In the context of **Inference of the Training Dataset**, the primary concern revolves around preserving the privacy of individuals who contribute data for the model's training. This aspect involves two key considerations: (1) *Membership inference*: Given a model and an instance, the adversary discriminates if the instance was in its training dataset. (2) *Property inference*: In cases where training data are distributed across different clients, the adversary tries to infer characters that are true of a subset of the training data.

In the realm of **Inference of Model Parameters**, diverging from the preceding aspects that center around data privacy, the primary focus is on safeguarding the privacy of the model itself. This aspect mainly encompasses *Model Extraction*. In this scenario, the adversary endeavors to acquire knowledge of an equivalent or closely approximated model by gaining access to the existing model, often through query access or other means.

4.2 Privacy Issue in Learning Across Different Domains

Cross-domain learning seeks to improve the learning of the target domain by leveraging knowledge acquired from the source domain. In today's landscape, increasing awareness and implementation of privacy protection measures limit access to diverse data sources, posing challenges for traditional cross-domain learning approaches. Consequently, there is a pressing need to advance cross-domain learning technologies that operate within stringent privacy constraints, ensuring alignment with evolving regulations. This adaptation is essential to address the changing landscape of data access and privacy concerns.

Existing research efforts in the realm of privacy within cross domains learning predominantly concentrate on safeguarding data privacy. Particularly, the concept of differential privacy, which

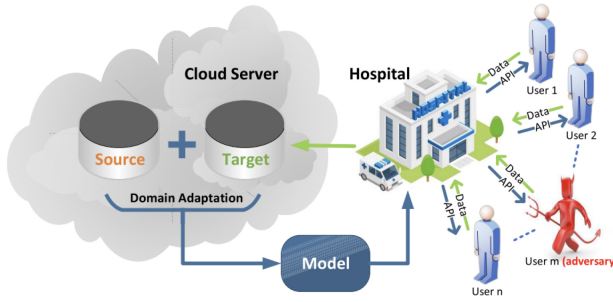


Fig. 12. Illustration of domain adaptation and the risk of privacy leakages. (Image from Reference [226].)

offers a mathematically rigorous framework for data privacy, is commonly employed as the guiding principle. Given a randomized algorithm M with an output space Ω and a well-defined probability distribution, it is considered (ϵ, δ) -differentially private if, for all pairs of adjacent datasets D and D' differing by a single data point d , and for all measurable sets $\omega \in \Omega$, the following inequality holds:

$$\Pr[M(D) \in \omega] \leq \exp(\epsilon) \Pr[M(D') \in \omega] + \delta.$$

Differential privacy necessitates that upon obtaining the output of a differentially private algorithm M applied to the confidential dataset D , there should be no means of ascertaining whether a specific data point d is present in the private dataset D or not. The parameter ϵ represents the privacy budget, signifying the extent of privacy inherent to the model. A diminished value corresponds to an augmented level of privacy.

4.3 Privacy-preserving Domain Adaptation

4.3.1 Privacy in Unsupervised Domain Adaptation. Domain adaptation endeavors to adjust a model trained in a source domain to a target domain. This adaptation occurs when the distributions of the source and target domains differ yet maintain a certain level of interrelation. In UDA scenario, data from the source domain possess labels while data from the target domain remain unlabeled. Contemporary UDA techniques proficiently convey knowledge from the labeled source domain to the unlabeled target domain. However, in practical contexts, as illustrated in Figure 12, where diverse institutions provide data for the source and target domains, the potential for privacy risks arises due to the exchange of information between these distinct domains.

While many previous works fail to take this aspect into consideration, a few recent works [7, 86, 106, 205, 226, 294] study how to perform unsupervised domain adaptation with privacy guarantees for training data. Le Tian et al. [106] solved the challenge of privacy-preserving domain adaptation with proposed **differentially private optimal transport (DPOT)** and **differentially private domain adaptation (DPDA)** model, as shown in Figure 13. DPOT uses the Johnson-Lindenstrauss transform [90] for data projection via a random matrix M to preserve pairwise distances in data. It computes the coupling matrix and Wasserstein distance between two domains with differential privacy by

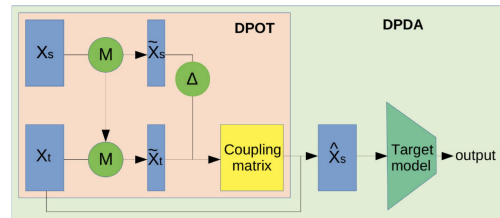


Fig. 13. The framework of Differentially Private Optimal Transport (DPOT) and Differentially Private Domain Adaptation (DPDA). (Image from Reference [106].)

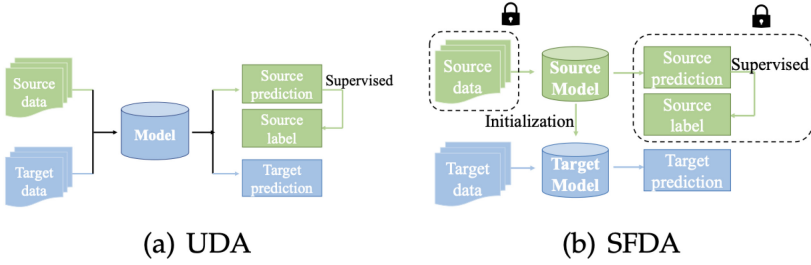


Fig. 14. Illustration of difference between UDA and SFDA. (Image from Reference [256].)

introducing noise to the random projection. DPDA, an extension of DPOT, leverages labeled source data to enhance a related target task while maintaining the privacy of each data source. Later, Wang et al. [226] introduced a global differentially private domain adaptation method for domain adaptation with privacy assurance using adversarial learning. They embedded differential privacy into specific layers and learning processes by incorporating Gaussian noise into the gradients. Different from Reference [226], Jin et al. [86] proposed a private correlation alignment approach for domain adaptation (PRIMA). In pursuit of privacy assurances, perturbations are introduced into the authentic covariance matrix, resulting in a confidential covariance matrix for the target domain. Moreover, during gradient-based model training, perturbations are applied to gradients to achieve a privacy-preserving model. Supplementary enhancements are implemented to address the constraints arising from the integration of these perturbation techniques.

Unlike the focus on privacy in unsupervised domain adaptation, Wu et al. [235] and Wang et al. [228] tackled privacy concerns within the context of a multi-source scenario, where multiple source domains are present. While existing efforts in data privacy primarily utilize differential privacy mechanisms in single-domain settings, they often neglect the distribution shifts across diverse domains. To counter this limitation, Wang et al. [228] introduced a methodology for differentially private multi-source hypothesis transfer learning. This technique assesses the interplay between source and target domains by utilizing hypotheses trained on the source domain. This strategy amplifies the refinement of target hypothesis acquisition by utilizing an accessible public unlabeled source dataset. Concurrently, Wu et al. [235] presented the MultiSTLP algorithm, which engages in multi-source selection transfer learning with an emphasis on privacy preservation. Distinct from Wang et al. [228], this approach leverages samples from a representative dataset culled from various sources, alongside unlabeled group probability samples within the target domain.

4.3.2 Privacy in Source-free Domain Adaptation. Common unsupervised domain adaptation algorithms typically involve access to source domain data. However, in cases where heightened data privacy is imperative—such as with sensitive data like medical records—sharing source data with the target domain becomes unfeasible. To relax the assumption regarding the availability of source data in existing domain adaptation methods, **Source-free Domain Adaptation (SFDA)** has been proposed in recent years. SFDA has emerged as a prominent topic in domain adaptation, garnering widespread attention in image classification, semantic segmentation, and object detection. Figure 14 illustrates the key distinctions between SFDA and UDA. In the UDA scenario, the model has access to both source and target domain data. However, in the SFDA setting, only the unlabeled target data and source domain pretrained model are available, which would make the UDA methods unsuitable for the SFDA scenario.

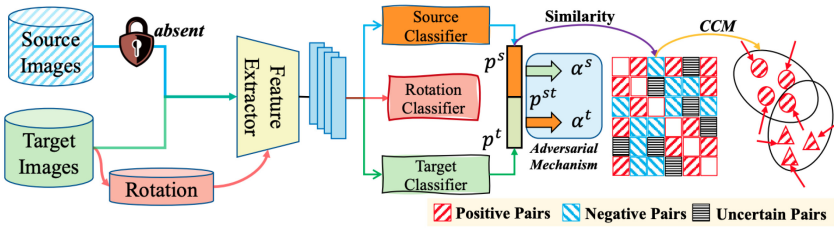


Fig. 15. Framework of the Adaptive Adversarial Network (A²Net). (Image from Reference [237].)

In the following, we offer a comprehensive review of privacy considerations in SFDA, examining it from two perspectives: data-based and model-based. Rather than providing an exhaustive review of every aspect, our goal is to highlight the primary research directions within these two domains.

Data-based methods can be roughly categorized into two types. First, addressing the unavailability of source domain data involves mimicking it using information from the model or reconstructing an alternative domain. Second, for unannotated target domain data, methods explore potential structures or clustering information within the unlabeled target domain, enabling independent domain adaptation relying solely on the target data. These are referred to as Domain-based Reconstruction and Image-based Information Extraction in our classification.

The core objective of domain-based reconstruction is to create a new domain that can serve as a surrogate for overseeing the target domain. Broadly speaking, three main strategies are employed to construct this new domain. First, one can generate a virtual source domain [104, 121, 213, 251], which attempts to construct a domain to compensate for the absence of source data. Second, there is the approach of forming a perturbed target domain for robust learning [29, 237, 296]. The final strategy involves segmenting the target domain into source-like and target-like segments [88, 108, 116], followed by the application of intra-domain adversarial techniques to achieve alignment. For example, Li et al. [121] proposed the Collaborative Class Conditional Generative Adversarial Networks to eliminate the dependence on source data. By integrating generated images into the model adaptation process, collaborative learning enhances both the prediction model and the generator. Xia et al. [237] introduced an Adaptive Adversarial Network, depicted in Figure 15, to acquire robust and discriminative representations through self-supervised learning, employing a rotation-based self-supervised technique on target data. Ding et al. [38] proposed Distribution Estimation to concentrate on estimating the class-conditioned feature distribution of the source domain by leveraging the target data and its corresponding anchor points.

The image-based information extraction methods can be broadly divided into two categories: those based on image semantic information [246–248] and those based on image style information [250, 278]. The former concentrates on the semantic information of target images, capitalizing on the observation that the target data inherently displays structured clustering. For example, Yang et al. [246] introduced **Attracting and Dispersing (AaD)**, the overall objective is described as

$$\mathcal{L} = \mathbb{E}[\mathcal{L}_i(C_i, \mathcal{B}_i)], \text{ with } \mathcal{L}_i(C_i, \mathcal{B}_i) = - \sum_{j \in C_i} p_i^T p_j + \lambda \sum_{m \in \mathcal{B}_i} p_i^T p_m, \quad (31)$$

where C_i is the K -nearest neighbors of i th target data, $\mathcal{B}_i$ contains the data that are not in C_i , and p is the output of the classifier. AaD promotes similar predictions for local neighborhood features in the feature space, while features that are farther apart in the feature space are expected to have dissimilar predictions. This approach facilitates efficient feature clustering and cluster assignment simultaneously. The latter investigates the style information of the image and transforms the target

image to the source style without altering the content information. This adaptation is intended to improve recognition by the source classifier.

Model-based methods are different from the data-based methods, which focus on data generation or exploring data properties. Model-based methods involve the separation of the model into distinct sub-modules. Within model-based methods, self-training emerges as the predominant paradigm and is widely acknowledged as the most popular approach in the broader SFDA research landscape. Lacking supervision for the source data, self-training utilizes the model's predictions on unlabeled target domain data to iteratively refine itself. It is often in conjunction with other techniques. The majority of self-training methods are implemented following the principles of pseudo-labeling [3, 23, 78, 127], entropy minimization [73, 101, 136], and contrastive learning [23, 77]. For example, Liang et al. [127] introduced **Source Hypothesis Transfer (SHOT)**, as shown in Figure 16, which utilizes information maximization and clustering-based pseudo-labeling to encourage the model for confident predictions on the target domain. Specifically, SHOT obtains the centroids for each class in target domain via weighted k-means clustering,

$$c_k = \frac{\sum_{x_t \in \mathcal{X}_t} \sigma_k(f_t(x_t)) g_t(x_t)}{\sum_{x_t \in \mathcal{X}_t} \sigma_k(f_t(x_t))}, \quad (32)$$

where f_t and g_t are classifier and feature extractor, respectively. $\sigma_k(f_t(x_t))$ denotes the probability of x_t belong to k th category. The refined pseudo labels are obtained via the nearest centroid classifier with the weighted centroids $\{c_k^i\}_{i=1}^K$. SHOT could update the centroids with Equation (32) and labels for multiple rounds. To make the target outputs individually certain and globally diverse, SHOT minimizes the following **information maximization (IM)** loss:

$$\mathcal{L}_{IM}(f_t; \mathcal{X}_t) = \underbrace{-\mathbb{E}_{x_t \in \mathcal{X}_t} \sum_{k=1}^K \sigma_k(f_t(x_t)) \log(\sigma_k(f_t(x_t)))}_{\text{Certain: } \mathcal{L}_{cer}} + \underbrace{\sum_{k=1}^K \hat{p}_k \log(\hat{p}_k)}_{\text{Diverse: } \mathcal{L}_{div}}, \quad (33)$$

where $\hat{p}$ is the mean output embedding of the whole target domain. The IM approach is effective in addressing the trivial solution scenario, wherein all unlabeled data shares the same one-hot encoding, through the fair diversity-promoting *Diverse*. The IM objective has proven to be a highly valuable function, widely adopted in numerous SFDA methods. Chen et al. [23] incorporated contrastive learning, pseudo-labeling, semi-supervised learning and IM into a unified framework, AdaContrast. As shown in Figure 17, the overall objective includes weak-strong consistency regularization $\mathcal{L}_t^{ce}$, the diversity regularization $\mathcal{L}_t^{div}$ in Equation (33), and the self-supervised contrastive learning regularization $\mathcal{L}_t^{ctr}$. Moreover, the effective identification of object regions plays a crucial role in domain adaptation [124]. Conventional CNN models often capture localized, domain-specific information, such as background details, which may not significantly contribute to highlighting the primary objects of interest. Consequently, the SFDA framework incorporates the self-attention mechanism to address this limitation, as discussed in References [13, 102].

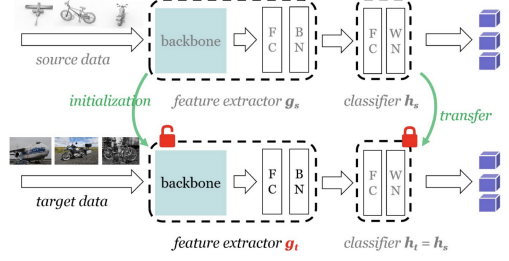
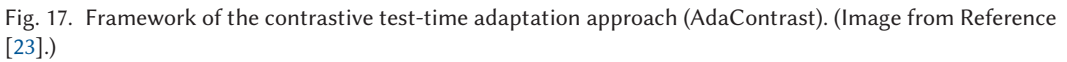


Fig. 16. Pipeline of the SHOT framework where the hypothesis remains frozen, and the feature learning module serves as the initialization for learning in the target domain. (Image from Reference [127].)



Different from domain adaptation, which assume the availability of target domain data, DG goes a step further by not requiring access to target data. The primary objective of DG is to learn a generalized model from one or multiple training domains with different probability distributions, enabling effective out-of-distribution generalization. This tutorial caters to machine learning researchers and industry practitioners, particularly those interested in transfer learning, domain adaptation, and generalization. The goal is to facilitate the learning and practical application of these techniques in real-world scenarios. The major concerns arise around the protection of the source data and model parameters.

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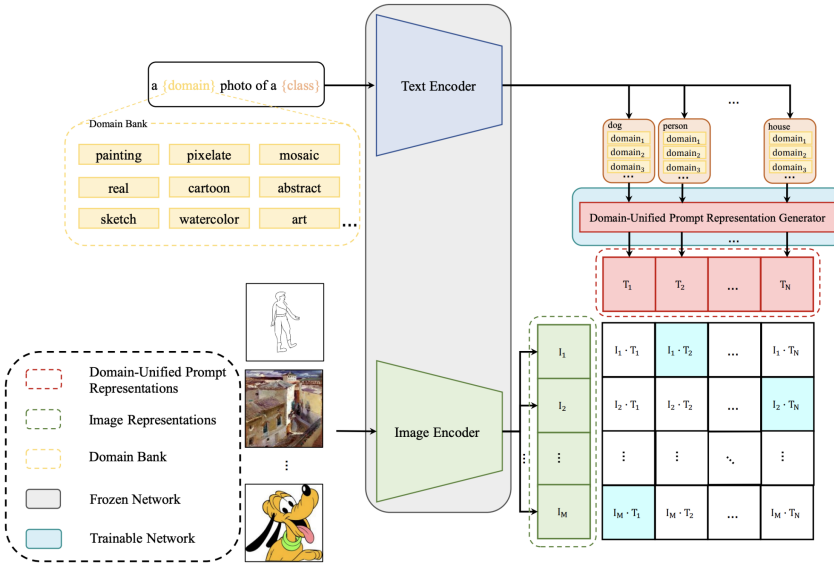


Fig. 18. Framework for source-free domain generalization. (Image from Reference [158].)

learnable style word vectors to generate a spectrum of style features. Notably, these style features are strategically positioned in proximity to their corresponding content features extracted from the text. The learned style word vectors are further employed in training a linear classifier.

4.5 Others

Federated Learning [93, 137, 149, 157, 165, 169, 176, 209, 234, 242, 265] adheres to data privacy constraints, typically disallowing data sharing between the server model and client models. A line of research [93, 137, 157, 165, 176, 209, 234, 242, 265] studied domain adaptation or domain generalization under the federated learning setting, where only parameter transmission is allowed. Different from traditional federated learning, it is more challenging, because datasets may differ in both the sample and feature space. In such a setting, transferring the knowledge safely without violating the privacy constraints is required. They are subject to the privacy constraint in federated learning setting, however, prohibiting data sharing is merely a basic approach to protect data privacy. For higher levels of data privacy protection, several recent works especially [54, 203, 261, 294] focus on the privacy protection in such setting. Different strategies are proposed for better adaptation under privacy constraints. To protect client data privacy, homomorphic encryption-based approaches [54, 203, 261] are proposed for effective knowledge transfer. More specifically, Zhang et al. [261] improves the commonly used maximum mean discrepancy with secure-aware technologies based on homomorphic encryption for effective knowledge transfer without compromising privacy. Song et al. [203] exploit homomorphic encryption to encrypt intermediate results, which need transmitting to protect privacy. Gao et al. [54] builds a federated transfer learning framework with a privacy preserving learning approach with two variants based on homomorphic encryption and secret sharing techniques. Zhuang et al. [294] further explore unsupervised domain adaptation within a federated learning context, with a primary emphasis on enhancing model performance through iterative aggregation of knowledge from the source domain via federated learning. In this approach, data privacy is safeguarded by exclusively transferring models, rather than raw data, between domains, omitting

additional privacy-preserving techniques. Kang et al., [93] study cross-domain federated learning problem with tabular datasets. Through extending the feature space via vertical federated with another feature-abundant party and tackling the sample-insufficient issue via adversarial domain adaptation from the additional party to the target party, they allow the secure and efficient privacy-preserving federated adversarial domain adaptation. Different from many existing adversarial domain adaptation models where low-interpretable feature representations with respect to the target task are typically learnt with a single feature extractor, Kang et al. [93] leverage domain expertise to partition the feature space into distinct groups, with each group encapsulating closely related features. Subsequently, they acquire semantically meaningful higher-order features from each designated group. Additionally, they also implement fine-grained domain adaptation for each feature group to enhance transferability. Different from other existing works, Kohen et al. [100] first presents a general framework for transfer learning in the hybrid-model.

4.6 Future Research Direction of Privacy

Privacy in Various Cross-domain Learning Scenarios. While the predominant emphasis in existing literature revolves around data privacy within unsupervised domain adaptation, it is imperative to extend our attention to privacy concerns in alternative domain adaptation and generalization tasks. These encompass scenarios such as semi-supervised domain adaptation, open-set domain adaptation, and open-set domain generalization. By addressing privacy challenges in a spectrum of cross-domain learning scenarios, we can foster a more comprehensive understanding and implementation of privacy-preserving techniques across various learning domains.

Expanding Source-free Domain Adaptation. Despite the significant surge in SFDA and the subsequent advancements in methodology and performance, there is a predominant focus on classification tasks within SFDA research, surpassing the progress made in other computer vision domains. Exploring several intriguing aspects could pave the way for potential directions to offer a more comprehensive understanding of SFDA and enhance its applicability across diverse scenarios. First, while pseudo-labeling and entropy minimization are widely employed in SFDA, introducing significant innovations would be beneficial for refining SFDA methodologies. Second, establishing universal theoretical foundations could address the limitation of existing SFDA methods that rely on intuitive motivations. Last, broadening the application of SFDA holds the potential to make a substantial contribution to unexplored privacy application settings and scenarios.

Various Dimensions of Privacy. In this survey, our primary emphasis is on exploring data privacy within the realm of representation learning across domains. However, other facets of privacy, such as those related to population characteristics and model parameters, remain relatively unexplored in the context of cross-domain learning. Numerous challenging issues persist, presenting opportunities for further exploration in the future.

5 FAIRNESS IN REPRESENTATION LEARNING ACROSS DOMAINS

Fairness in machine learning [65, 202, 273] is a burgeoning area of interest. It is dedicated to safeguarding the interests of individuals and groups within decision-making or recommendation systems. The application of fairness across domains adds a layer of realism but also poses increased challenges. This section delves into the concept of the unfairness circle and its sources in cross-domain scenarios. The focus is primarily on situations where fairness is sought in the train/test phase under cross domains situation, as this differs from scenarios such as class incremental continual learning, which align more closely with traditional fairness settings. Various approaches, including bounding-based, training-based, and causal inference-based methods, are discussed for achieving fairness. The section concludes by highlighting current challenges and suggesting possible future directions.

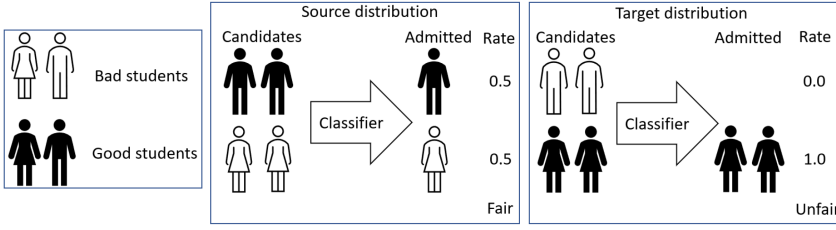


Fig. 19. Example for **unfairness in covariate shift**. It shows the same prediction model for admission in different data distribution environments. In source domain, to satisfy the same admission rate between males and females, the classifier will give the higher weight on gender attributes. However, in target domain, the model preference would cause the unfairness that female admission rate is higher than male's.

5.1 Unfairness Circle and Source of Unfairness Across Domains

For transitional unfairness issues, the original bias emerges from imbalanced data, algorithms and user interactions. It is almost invariant in cross domain area. For data perspective, imbalanced data in source domain first introduce unfairness. For example, the dominant attribute classes would play more important role in prediction while ignore minor classes. In addition, though it satisfies great in source domain, it may cause serious unfairness when deploying in real world due to distribution shift. However, algorithms aiming to fit the distribution of the source domain may inevitably distort the distribution in the target domain, resulting in less accurate and fair predictions [27, 31, 177]. Moreover, the prediction result would further influence user preferences. The behavioural biases will aggregate the unfairness in data.

Unfairness issues in cross domain areas are mainly result from distribution shift [5] or domain shift [208]. Distribution shift means that data distribution between training dataset and real world situation is different. In general, distribution shift can be classified as two types, including **covariate shift** [62, 156, 196, 207] and **label shift** [9, 129, 188]. Covariate shift means the disparity of input variables distribution between different datasets. It will raise unfairness issues. For example, Figure 19 shows a classifier is fair in source distribution while it would produce unfairness results in target distribution, since it would give higher weights on female attributes. As for the label shift, it means that feature distribution remains the same, the label distribution might changes. For example, one school may admit students with grades of B or higher, but another school will only admit students with grades of A. It is obvious that a fair model training on one school dataset will not be fair on the other school. However, a lot of works [27, 31, 160, 177, 202] try to mitigate such unfairness and improve the fairness transferability.

Typically, fairness in representation learning spans across various training stages, and they can also be categorized into pre-process, in-process, and post-process methods. Each stage addresses specific aspects of fairness, aiming to mitigate biases and ensure equitable outcomes. In the pre-process stage, methods focus on preprocessing data to identify and mitigate bias before the model is trained. In-process methods intervene during the training process to promote fairness and reduce disparate impact. Post-process methods, however, address fairness concerns after the model has been trained, aiming to adjust predictions or outcomes. To provide a comprehensive overview, we have summarized the relevant papers in Table 6.

5.2 Fairness in Domain Adaptation

As previously mentioned, the domain adaptation may have different distribution between source and target datasets, which makes the fairness more challenging. Typically, with different training stages, fairness can be divided into pre-process method, in-process method, and post-process

Table 6. Summary of the Typical Fairness in Representation Learning Span Various Scenarios

Methods	Goals	Fairness metrics	Phases	Cross Domain Conditions
[227]	Data	APG	Pre-process	Domain adaptation
[27]	Bound	DP, EO	In-process	Domain adaptation
[6]	Prediction	EO, APG	In-process	Domain adaptation
[202]	Prediction	DP, EO, CF	In-process	Domain adaptation
[146]	Prediction	DP, EO	In-process	Domain generalization
[272]	Prediction	DP	In-process	Domain generalization
[192]	Prediction	AIF	In-process	Domain generalization
[214]	Segmentation	EO	In-process	Domain adaptation
[160]	Representation	DP	In-process	Domain adaptation
[200]	Prediction	DP, EO	Post-process	—

Goal refers to the target task. Specifically, “Data” means making training data unbiased; “Bound” means setting up a fairness upper bound without downstream task; “Prediction” is predicting labels; and “Representation” refers to generating fair latent embeddings. In the “Fairness metric” column, DP denotes demographic parity [44] or statistic parity, “EO” refers to Equal opportunity [70], “CF” means counterfactual fairness [105], “DBC” is Decision Boundary Covariance [258], and “APG” [270] denotes the Accuracy Parity Gap. “AIF” [192] represents the Average Individual Fairness. “Phase” denotes the debiasing stage of a method.

method. Pre-processing methods mitigate unfairness by modifying training data, such as data augmentation. For example, Wang et al. [227] tried to deal with fair domain adaptation classification problem. They generate supplementary samples in the source domain for few-shot classes to improve classification accuracy for minor class. However, in across domain area, there is not much work in the pre-processing category, since it is intractable to include all possible data distribution during training process by modifying data. In processing approaches, the methods can be broadly categorized into three main groups: **bounding-based, causal inference-based, and training-based**.

5.2.1 Bounding-based Approaches. The predominant and intuitive methodologies in addressing fairness concerns are bounding-based approaches [6, 27, 31, 189, 214, 252, 270, 271]. These approaches establish fairness bounds, such as fairness definitions, to regulate representations or downstream tasks. The general loss function is defined as

$$\mathcal{L} = \mathcal{L}_t + \mathcal{L}_f + \mathcal{L}_{da}, \quad (34)$$

where $\mathcal{L}$ is the typical loss function, $\mathcal{L}_t$ is the loss for the target task, $\mathcal{L}_f$ represents the fairness loss regulated by any bounds, and $\mathcal{L}_{da}$ is the domain adaptation loss. For example, as shown in Figure 20, Schumann et al. [189] introduced a general framework that contains task head $g(\cdot)$, fairness head $a(\cdot)$, and transfer head $d(\cdot)$ to optimize Equation (34) via the following objective function:

$$\min \left[\sum_{(Z, Y) \in \mathcal{D}_{S+T}} \mathcal{L}_t(Y, g(h(Z))) + \sum_{(A, Z^0) \in \mathcal{D}_{S^0}} \lambda_f \mathcal{L}_f(a(h(Z^0)), A) + \sum_{(D, Z^0) \in \mathcal{D}_{S^0+T^0}} \lambda_{da} \mathcal{L}_{ad}(d(h(Z^0)), D) \right], \quad (35)$$

where Z denotes the feature, Y represents the ground truth label, A is the attribute, D is the domain label, $\mathcal{D}_{S^0+T^0}$ stands for the sensitive attribute from the source/target domains, $\mathcal{L}_t$ is the cross-entropy loss, and $\mathcal{L}_f$ and $\mathcal{L}_{da}$ are MMD loss [139]. The second term in Equation (35) aims

to regularize the sensitive attributes in the source domain, while the third term in Equation (35) aims to regularize the source and target memberships. Later, Yoon et al. [252] redefined the fairness metric using Wasserstein distance and incorporated it into domain adaptation techniques, also leveraging Wasserstein distance. Bang et al. [6] introduced a simulation for equitable label propagation, ensuring model consistency across different groups, and proposed a theoretical upper bound for **fair consistency regularization (FCR)**. Chen et al. [27] developed a **fairness transferability subject (FTS)**, which is the general upper bound for transferred

group fairness violation, accounting for distribution shift, including covariate shift and label shift.

In contrast to the aforementioned studies, Coston et al. [31] tackled a more challenging scenario where sensitive attributes are available in either the source or target domain. They employed demographic parity as their fairness metric and utilized weighting methods to minimize empirical risk, narrowing the gap in group-specific prevalence. Similarly, Zhang et al. [270] addressed fairness in incomplete data problems, where some attribute values are missing. Zhao et al. [271] applied fairness in online meta-learning, using decision boundary covariance to regulate the model. Truong et al. [214] incorporated fairness into the image segmentation task by adapting Equalized Odd limitation to segmentation classes.

5.2.2 Causal Inference-based Approaches.

Causal inference methods utilize causal information, particularly in addressing unfairness issues, especially within the context of counterfactual fairness. We can represent the problem and fairness objectives through a common causal graph, as illustrated in Figure 21. Here, A , R , T , and Y are system variables. A represents the sensitive attribute, and Y is the label for prediction. C is the context variable, indicating the domain distribution. In other words, $P(A, R, T, Y|C = 0)$ and $P(A, R, T, Y|C = 1)$ represent the joint distribution for the source and target domains, respectively. The fairness problem can now be formulated as a minimization problem, i.e.,

$$f_{target}^* = \arg \min_{f \in F} \{ (P(f(S) \neq Y|C = 1) + G(f, P(Y, S|C = 1))) \leq \epsilon \}, \quad (36)$$

where $S = A, R, T$ is the input features, and $G(f, P_{target})$ is the fairness constraint, and f is the learnable function and ϵ is the hyperparameter controlling the intensity of constraints. In this equation, we have two terms. $P(f(S) \neq Y|C = 1)$ describes the classification error. Minimizing it would result in higher accuracy. $G(f, P(Y, S|C = 1))$ is the fairness constraint in target domain. Lower value indicates higher fairness achievement. Singh et al. [202] proposed **fairness violations and mitigation (FVM)** to address the challenge of an unseen target distribution. They utilized causal information to model data shifts and leverage conditional independencies in the data for

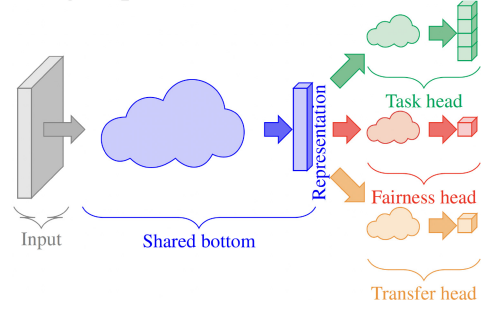


Fig. 20. General framework combines a primary training objective, a fairness objective, and a transfer objective to improve fairness goals in a target domain. (Image from Reference [189].)

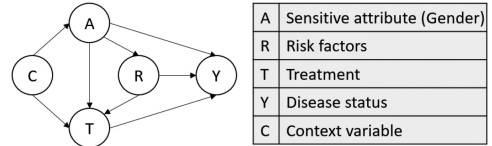


Fig. 21. Causal graph for a disease treatment example. A, R, T, Y are system variables. C is the context variable. Arrow indicates the causality. For example, $T \rightarrow Y$ indicates that treatment will influence disease status.

estimating both model accuracy and fairness metrics. Their approach encompasses group-fairness metrics, such as demographic parity and Equalized Odds, extending to counterfactual fairness. In a similar vein, Zhao et al. [273] inferred decision boundary covariance from fair causality and causal effect. They ensured fairness by controlling the covariance between protected variables and the signed distance from the feature vectors to the decision boundary.

5.2.3 Training-based Approaches. Training-based methods [74, 160, 177, 206] aim to construct a training strategy to achieve fairness. For example, adversarial training strategy trains a fair representation to fool a sensitive discriminator. If the sensitive discriminator cannot predict sensitive information, then the representation and its downstream task can achieve fairness. Stan et al. [206] converted fair prediction problem to a competition between an adversary to fairly minimize predictive loss and an adversary to maximize that objective loss. Figure 22 describes the proposed framework. Specifically, they constructed the fairness loss, denoted as $\mathcal{L}_{\text{fair}}$, which defined by the binary cross-entropy loss applied to the predictions of the attribute classifier h_w composed with the encoder e_u on the data x with attributes a :

$$\mathcal{L}_{\text{fair}} = \mathcal{L}_{\text{bce}}((h_w \circ e_u)(x, a), a). \quad (37)$$

They proposed a two-step process for ensuring fairness: (1) Keep the encoder e_u constant and minimize the fairness loss $\mathcal{L}_{\text{fair}}$ by updating the weights of the attribute classifier h_w . (2) With the attribute classifier h_w fixed, maximize the fairness loss $\mathcal{L}_{\text{fair}}$ by adjusting the encoder e_u . The total loss is expressed as follows:

$$\mathcal{L} = \mathcal{L}_{\text{bce}}(\hat{y}, y_{\text{src}}) + \alpha \mathcal{L}_{\text{fair}}^{\text{src}} + \beta \mathcal{L}_{\text{fair}}^{\text{tar}} + \gamma \mathcal{L}_{\text{swd}}, \quad (38)$$

where $\mathcal{L}_{\text{bce}}$ is the target loss function to train the classifier with labeled source domain, $\mathcal{L}_{\text{swd}}$ represents the Sliced Wasserstein Distance loss to align source and target domain via optimal transport, and $\mathcal{L}_{\text{fair}}^{\text{src}/\text{tar}}$ denotes Equation (37) with source/target data. By applying the loss to both source and target domain, it can achieve fairness across domains.

Rezaei et al. [177] proposed an approach, **robust fairness under covariate shift (RFCS)**, for obtaining a predictor that is robust to covariate shift and satisfies target fairness requirements. They introduced a penalty term to evaluate fairness in the target distribution and optimized the problem accordingly. Hong et al. [74] addressed federated learning situations, ensuring privacy by not accessing users' sensitive group information during the debiasing process. Users can choose to participate in the adversarial component, giving them control over privacy and computational concerns. Additionally, a multi-task training strategy is employed to formulate fair and robust representations. Oneto [160] focused on generating robust and fair representations. They achieved this by learning fair and transferable representations through multi-task learning, ensuring guarantees on generalization to the target domain.

5.3 Fairness in Domain Generalization

DG sets itself apart from domain adaptation by not assuming access to target domain data. The primary objective of DG is to build a model with robust generalization capabilities to new domains. It learns from one or several source domains with diverse probability distributions, facilitating

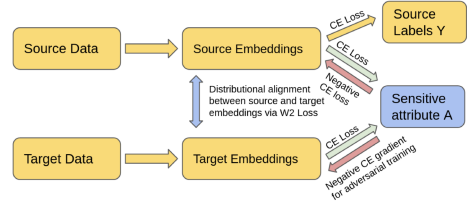


Fig. 22. Block-diagram description of the proposed framework. (Image from Reference [206].)

effective adaptation to out-of-distribution scenarios. Introducing fairness considerations in DG poses challenges due to the absence of target domain samples. For instance, the adversarial training-based method proposed by Reference [206] requires target data for adversarial training, aiming to mitigate unfair biases in both representation and classifier. Without access to target samples, implementing such a process becomes unfeasible.

Nevertheless, there are methods available for transferring fairness in domain generalization. Sharifi et al. [192] present a bounding-based approach that concentrated on individual fairness. They employ a twofold constrained optimization approach, simplifying the fair learning task into a strategic interaction resembling a game with two roles: a “Learner” (the primal participant) and an “Auditor” (the dual participant). Mandal et al. [146] is a training-based approach, **ensuring fairness (EF)**, which trained a fair classifier that remains robust to any distributions, representing arbitrary weighted mixtures of training distributions. They formulated a min-max objective function. On the one hand, it minimizes distributionally robust training loss, and, on the other hand, it seeks a fair classifier within a specified class of distributions. As shown in Figure 23, Pham et al. [167] not only established theoretical bounds on unfairness and expected loss at deployment but also introduced the **Fairness and Accuracy Transfer by Density Matching (FATDM)** framework to address fairness in domain generalization. They developed upper bounds for expected loss and unfairness in target domains, along with a lower bound for expected loss. Consider a special case where the unfairness measure is defined as the distance between means of two distributions:

$$\epsilon_D^{EO}(\hat{f}) = \sum_{y \in \{0,1\}} \left| \mathbb{E}_D \left[\hat{f}(X)_1 | Y = y, A = 0 \right] - \mathbb{E}_D \left[\hat{f}(X)_1 | Y = y, A = 1 \right] \right|. \quad (39)$$

Then the unfairness at any unseen target domain D^T is upper bounded:

$$\begin{aligned} \epsilon_{D^T}^{EO}(\hat{f}) &\leq \frac{1}{N} \sum_{i=1}^N \epsilon_{D_i^S}^{EO}(\hat{f}) + \sqrt{2} \min_{i \in [N]} \sum_{y \in \{0,1\}} \sum_{a \in \{0,1\}} d_{JS} \left(P_{D^T}^{X|Y=y, A=a}, P_{D_i^S}^{X|Y=y, A=a} \right) \\ &\quad + \sqrt{2} \max_{i,j \in [N]} \sum_{y \in \{0,1\}} \sum_{a \in \{0,1\}} d_{JS} \left(P_{D_i^S}^{Z|Y=y, A=a}, P_{D_j^S}^{Z|Y=y, A=a} \right). \end{aligned} \quad (40)$$

The upper limit specified in Equation (40) comprises three components, with the second component being unmanageable during training as it is contingent upon the unobserved target domain and is rooted in the input space. To secure fairness within the target domain D^T , the representation mapping g and the model $\hat{h}$ are trained to maintain invariant $P_{D_i^S}^{Z|Y,A}$ across the source domains.

Concurrently, the composite function $\hat{f} = \hat{h} \circ g$ is optimized to achieve considerable fairness in the source domains. Moreover, they demonstrated that fairness and accuracy can be perfectly transferred from source domains to target domains through invariant representation learning under certain identified sufficient conditions. Consider N source domains $\{D_i^S\}_{i=1}^N$ and an unseen target

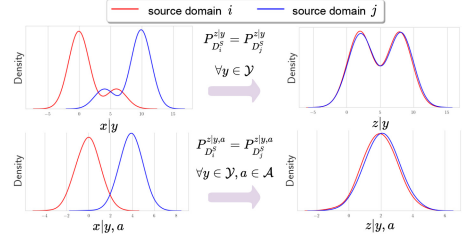


Fig. 23. Depiction of domain-invariant representation: To ensure that both accuracy and fairness are effectively transferred to target domains, it is necessary to identify a representation, denoted as z , that satisfies the condition where both $P_{D_i^S}^{z|y}$ and $P_{D_i^S}^{z|y,a}$ remain consistent across different domains. (Image from Reference [167]).

domain D^T . Define set $\Lambda = \{D^t : D^t = \sum_{i=1}^N \pi_i D_i^S, \{\pi_i\} \in \Delta_{N-1}\}$. To transfer fairness in domain generalization, $\forall D^T \in \Lambda$, if g is the mapping under which $P_{D_i^S}^{Z|Y,A}$ is the same across all source domains, then

$$\epsilon_{D_i^S}^{\text{EO}}(\hat{h}) = \epsilon_{D^T}^{\text{EO}}(\hat{h}) = \epsilon_{D_i^S}^{\text{EO}}(\hat{f}) = \epsilon_{D^T}^{\text{EO}}(\hat{f}), \forall i. \quad (41)$$

The FATDM framework consists of a two-stage training process involving density matching function learning and domain-invariant representation learning. Zhao et al. [272] introduced a novel approach, the first to integrate fairness considerations within a framework adept at handling covariate and dependence shifts. They redefined the problem as a constrained learning issue, establishing duality gap bounds and introducing a unique upper bound tailored for fairness in the target domain. Their method not only maintains fairness but also ensures robust representations against distributional shifts, common challenges in domain generalization.

5.4 Future Research Direction of Fairness

Fairness Definition and Evaluation. The fairness in across domain area is still a emerging topic. The definition of fairness mostly rely on traditional metrics, i.e., demographic parity [44], equal odds [70]. However, most work currently focuses on a fixed target domain. In realistic, target domain may varied and even changes every time. Thus, how to define a general fairness to cover the online situation. In addition, in some tasks, such as face recognition, it is important to achieve fairness in both source and target domains. How do we define such fairness that can cover both domains?

Interpretable Fairness. Fairness models are working on achieve fairness in mathematical or statistical way no matter in independent and identically distributed data or across domain data. How can we use some explainable way, such as causal graph or saliency map, to interpret models so that it can further direct humans to make non-discriminatory judgments?

Fairness in Foundation Models. Foundation models such as the SAM- [98] and CLIP-based models exhibit remarkable zero-shot capabilities. However, their effectiveness is often hindered by inherent biases and imbalances present in their training data, leading to challenges in ensuring fairness across various contexts. Several recent studies [190, 233] have diligently quantified the extent of unfairness exhibited by these multi-modal models. Addressing these fairness concerns requires innovative research directions that focus on enhancing the fairness of foundation models for downstream tasks, even in the face of domain shifts. A promising avenue involves strategies to mitigate these issues. For instance, one approach is through prompt learning. This technique encompasses the design of carefully crafted prompts or the acquisition of dynamically adaptable soft prompts, enabling the foundation model to consciously reduce attention toward sensitive attributes. This adjustment can significantly contribute to improving the fairness performance of the model.

6 EXPLAINABILITY IN REPRESENTATION LEARNING ACROSS DOMAINS

Interpretable and explainable machine learning models emerge from a situation that human is model agnostic. To understand why models work well or not, interpretable and explainable models are proposed to help explain machine learning models in an intuitional way. Explainability has been studied in many areas. It can be defined as “the ability to explain or to present in understandable terms to a human” [39]. In the context of representation learning across domains, we hope we can know what a model learns from source domain and how a model will apply it to target domain. Thus, when a model is deployed in real world that work poor or violate some safety [71] or ethnicity [179] rules, we can easily troubleshoot them.

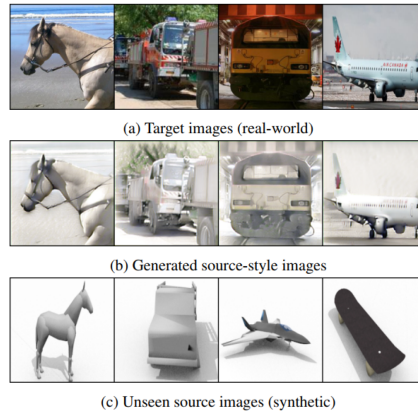


Fig. 24. Visualization of adapted knowledge in unsupervised domain adaptation on the VisDA dataset [166]. Panel (a) shows the target images. Panel (b) shows images generated from panel (a). Panel (c) presents the source images, which is inaccessible during generating process. (Images from Reference [75].)

In this section, we first introduce the goal of explainability models and some concepts, and then we summarise current work in explainable and interpretable work in across domain area. We divide them into explainable models and interpretable models. Finally, we will talk about its future directions.

6.1 Concepts

6.1.1 Interpretability. Interpretability requires models to give accurate and meaningful outputs, i.e., it not only answers what the results are but also explains why the results are presented. Specifically, interpretable models reflect physical constraints, monotonicity, additivity, causality, sparsity, and/or other desirable properties [18, 147, 180, 187]. In the context of representation in across domains, the interpretability of models leverage intrinsically interpretable designs to explain predictions or generations.

6.1.2 Explainability. Explainability refers to explain an existing black-box model post hoc. Explainable model is independent of target model, which we try to explain. Basically, it aims to analyze the results that a model gives in a textual, visual or symbolic way [130]. It can be model agnostic or model specific. In either way, it should satisfy Faithfulness, Consistency, Comprehensibility, Certainty, and Representativeness [18].

6.2 Explainability in Domain Adaptation

Explainability in DA and DG are similar in most cases. However, once the explainable method requires the target domain samples, it can only apply to domain adaptation methods. The key is to use existing training samples and relations to explain other samples. For example, explainable models [75, 173] aim to explain black box prediction models. Hou et al. [75] induced a **source-free image translation (SFIT)** approach, which is model-agnostic one. SFIT visualizes the adapted knowledge difference with image translation. As shown in Figure 24, they portray the knowledge learned from different UDA models and find that these generated images are able to capture the difference between source and target domains. Raab et al. [173] introduce the Domain Adversarial Tangent Subspace Alignment network. This framework conceptualizes data in the form of affine subspaces and employs adversarial techniques to align local manifold approximations between distinct domains. Kamakshi et al. [92] introduced XSDA-Net, an instance-level interpretable model

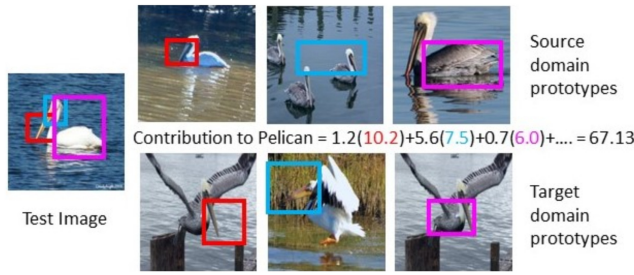


Fig. 25. XSDA-Net instance-level explanation. It extract some image prototypes from source domain and target domain. According to comparing test images with these prototypes, the similarity scores are forwarded to prediction. (Image from Reference [92].)

capable of providing explanations for each sample. XSDA-Net elucidates the prediction of every test instance by referencing prototypes in both the source and target samples. It extracts prototypes and calculates similarity with these prototypes, as illustrated in Figure 25.

6.3 Explainability in Domain Generalization

Explainable methods that do not rely on target samples can be applied to domain generalization models. These methods often reflect the general saliency map and require outside knowledge. Zunino et al. [297] explains their across domain classification network by the **explainable deep classification models (EDCM)**, which utilizes the means of a saliency map conveying how much each pixel contributed to the network prediction. Though saliency map is wildly used in interpretable models, it has some limitations. For example, saliency map is unable to give sufficient explanations for those false predictions, since it focuses on the same pixels when it predicts correctly.

Other across domain generating task also involves interpretability, such as semantic autoencoder, GAN. Back from Reference [99], Kodirov et al. tried to align latent feature space with semantic feature space. In this way, image features can be translated into semantic level, vice versa. Conditional GAN [152] also wants to generate images while controlling some semantic information. Gao et al. [55] used variational autoencoder and GAN jointly to generate unseen features. Li et al. [117] introduced LisGAN, a method capable of directly generating novel features from random noise conditioned by semantic descriptions. These generative techniques inherently possess explainability.

6.4 Future Research Direction of Explainability

Explainable Metrics. Explainability of existing models are unquantifiable. For example, these works [173, 297] use saliency map to explain their models. However, it is intractable to prove whether they give great explanations or not. Thus, a metric for explainability is needed to measure those explainable models.

High-level explanations. No matter, interpretable models [92, 297] or explainable models [75] explain across domain models in a low-level way, such as saliency map. Although it can bring some latent information for us to understand a model, it is still unable to troubleshoot misclassified samples. Therefore, high-level explanations, like nature language and semantic information, are necessary to better assist humans to understand models.

7 CONCLUSION

In this survey, we instantiate the trustworthy AI core concepts to representation learning across domains. We introduce four foundational crucial concepts of a trustworthy AI system, i.e., robustness, privacy, fairness, and explainability, to build our trustworthy representation learning across

domains framework. For each concept, we present a comprehensive overview of related methodologies to help the reader understand how each concept is addressed. Additionally, we provide comprehensive details of advanced technologies to ensure a thorough understanding of the research. Furthermore, we delve into potential future research directions related to each concept, aiming to inspire further advancements in trustworthy representation learning across domains.

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